

Benchmarking Grover–Rudolph State Preparation for Utility-Scale Quantum Numerical Integration

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Abstract

This report documents the methodology, implementation, and experimental results of the IBM Quantum Credits project on benchmarking Grover–Rudolph quantum state preparation for amplitude-estimation-based numerical integration on IBM quantum hardware.

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1 Introduction

This report consolidates the first experimental phase of the IBM Quantum Credits project devoted to benchmarking Grover–Rudolph state preparation for amplitude-estimation-based numerical integration on IBM quantum hardware.

The practical question addressed in this phase is not whether asymptotic quadratic speedups are visible on present-day devices. Rather, the objective is to establish a reproducible methodology for deciding which amplitude-estimation circuits are credible candidates for hardware execution, how compilation choices affect their observed accuracy, and how small structured benchmarks can be used to expose failure modes before larger Grover–Rudolph instances are attempted.

The campaign follows the feedback received with the IBM Quantum Credits award. In particular, it combines a QAE-oriented benchmark perspective [7] with Benchpress-style attention to circuit construction, transpilation, depth, and two-qubit-gate metrics [8]. It also builds on the project’s own theoretical foundation: a rigorous Grover–Rudolph state-preparation analysis [3], an angle-structure characterization of quantum numerical integration [1], an intrinsic gate/compilation viewpoint based on Lie group embeddings [2], and the broader algebraic probability framework in which quantum computation is naturally formulated [4]. The initial

hardware target is IBM’s Heron backend `ibm_fez`. The first accepted result set demonstrates a complete evidence chain:

$$g_0 \text{ calibration} \longrightarrow g_1 \text{ affine stress test} \longrightarrow g_2 \text{ quadratic stress test}.$$

All accepted hardware estimates are produced with a reproducible audit trail, including logical circuits, transpiled QASM, backend metadata, job identifiers, counts, and post-processing tables.

2 Background

Let A be a state-preparation circuit such that

$$A|0\rangle = \sqrt{1-a}|\psi_0\rangle|0\rangle + \sqrt{a}|\psi_1\rangle|1\rangle,$$

where the last qubit marks the event of interest and $a \in [0, 1]$ is the amplitude to be estimated. For the amplification level k , the ideal Grover-amplified success probability is

$$p_k(a) = \sin^2((2k+1) \arcsin \sqrt{a}).$$

Maximum-likelihood amplitude estimation (MLAE) estimates a by fitting the observed counts across a schedule of amplification levels K , avoiding the full quantum phase estimation register used in canonical QAE.

Grover–Rudolph state preparation [6] is relevant for numerical integration because efficiently integrable probability densities can be encoded into quantum amplitudes through controlled rotations. The rigorous version used as a reference point in this project formalizes the dyadic probability tree, proves exact preparation of the target measurement law, and gives perturbation and sampling design rules for finite-precision rotations [3]. In this project, the first hardware campaign uses the small SISC calibration family g_0, g_1, g_2 , which is rich enough to test constant, affine, and quadratic angle encodings while still allowing exhaustive local auditing before hardware submission.

2.1 Dyadic grid notation

For n grid qubits, set

$$B_n = \{0, 1\}^n, \quad N = 2^n.$$

The bit string $b = (b_0, \dots, b_{n-1}) \in B_n$ labels the integer

$$i(b) = \sum_{j=0}^{n-1} b_j 2^{n-1-j} \in \{0, \dots, N-1\}.$$

The midpoint dyadic grid is

$$D_n^M = \left\{ x_i^M = \frac{i+1/2}{2^n} : i = 0, \dots, 2^n - 1 \right\}.$$

Unless a quadrature rule is explicitly stated, the shorter notation D_n means D_n^M throughout the Grover–Rudolph probability-law experiments. For quadrature-rule QAE experiments we keep the rule visible and write, for example,

$$x_i^L = \frac{i}{2^n}, \quad x_i^M = \frac{i+1/2}{2^n}, \quad x_i^R = \frac{i+1}{2^n}.$$

The Simpson value is not a new grid D_n ; in this report it denotes the standard weighted combination of left, midpoint, and right evaluations used in the corresponding quadrature script. This convention prevents a collision between the support of a finite probability law and the quadrature rule used to approximate a continuum integral.

2.2 Angle-structure hierarchy for encoded functions

The encoding-complexity paper [1] introduces a hierarchy of classes that is directly relevant to the present experiments. For an n -qubit grid function $g_n : B_n \rightarrow [0, 1]$, usually obtained by sampling a continuum function g as $g_n(b) = g(x_{i(b)}^M)$, define the angle map

$$\Theta_{g,n}(b) = 2 \arcsin \sqrt{g_n(b)} = 2 \arcsin \sqrt{g(x_{i(b)}^M)}, \quad b \in B_n.$$

The sampled function belongs to $\mathcal{G}_n^{(d)}$ when $\Theta_{g,n}$ has a multilinear expansion in the bits $b_0, \dots, b_{n-1}$ of degree at most d :

$$\Theta_{g,n}(b) = \sum_{S \subseteq \{0, \dots, n-1\}} \hat{\Theta}_S \prod_{j \in S} b_j, \quad \hat{\Theta}_S = 0 \quad \text{for } |S| > d.$$

This degree is not the ordinary polynomial degree of $g(x)$. It is the degree of the rotation-angle structure seen by the amplitude oracle. Consequently, smoothness or a low-degree formula in x does not by itself imply a low-complexity encoding. The calibration functions used at the start of this project are the first three nontrivial levels of this hierarchy on the two-grid-qubit midpoint grid:

$$g_0 \in \mathcal{G}_2^{(0)}, \quad g_1 \in \mathcal{G}_2^{(1)}, \quad g_2 \in \mathcal{G}_2^{(2)}.$$

In the software this classification is checked by applying the finite difference/Walsh–Hadamard transform to the sampled angle map. The same diagnostic is used below to label later test functions whenever they are interpreted as amplitude-oracle functions.

2.3 Angle structure and quantum integrability

The angle hierarchy is not only a naming convention for benchmark functions; it is a natural candidate for discussing when a finite integration problem is pertinent for a quantum implementation. In an amplitude-estimation workflow, a test function $g_n : B_n \rightarrow [0, 1]$ is not encoded through its values directly, but through the rotation-angle map

$$\Theta_{g,n}(b) = 2 \arcsin \sqrt{g_n(b)}.$$

If $\Theta_{g,n} \in \mathcal{G}_n^{(d)}$ with small d , then the angle oracle has low multilinear interaction order in the grid bits. This does not by itself prove a hardware advantage, but it is strong structural evidence that the amplitude-encoding layer is compatible with short controlled-rotation decompositions. Conversely, a function that is elementary as a formula in x can still induce a full-degree angle map after the nonlinear $2 \arcsin \sqrt{\cdot}$ transformation and restriction to D_n , making it much less suitable for amplified QAE on current hardware.

This report therefore separates two related notions of “efficiently integrable”:

- *QAE-efficient integration*, where the quantity to be integrated is itself loaded as an amplitude function. Here the class $\mathcal{G}_n^{(d)}$ is directly relevant because it controls the structure of the angle oracle.
- *Grover–Rudolph distribution integration*, where a probability law p is first prepared and classical test functions f are integrated by postprocessing measured frequencies, $I_p(f) = \sum_i p_i f(x_i)$. Here the relevant quantum cost is the probability-tree preparation of p , while the angle class of f remains useful as a diagnostic for deciding whether the same quantity should later be attempted through amplified QAE.

The working hypothesis suggested by the present experiments is that low angle-degree functions are the appropriate first candidates for amplified QAE, whereas high angle-degree or localized quantities should first be studied through direct Grover–Rudolph distribution sampling. The hypothesis is deliberately conditional: actual efficiency also depends on the dyadic probability law, grid size, compilation, two-qubit depth, mitigation strategy, and the classical MC/QMC baselines used for comparison.

2.4 Finite algebraic probability model

All Grover–Rudolph cases in this report are finite probability models on a dyadic grid. For n grid qubits we use the midpoint grid defined in Section 2.1,

$$D_n = D_n^M = \left\{ x_i = \frac{i + 1/2}{2^n} : i = 0, \dots, 2^n - 1 \right\}.$$

Following the algebraic-probability viewpoint of [4], the finite classical model is the commutative algebra of grid functions $\mathcal{A}_n \simeq \mathbb{C}^N$. Its primitive events are the idempotents e_i , where $e_i(x_j) = \delta_{ij}$. A probability vector $p = (p_0, \dots, p_{N-1})$, with $p_i \geq 0$ and $\sum_i p_i = 1$, defines the state

$$\varphi_p(f) = \sum_{i=0}^{N-1} p_i f(x_i), \quad f \in \mathcal{A}_n.$$

Thus a numerical integral, payoff, event probability, or moment is treated as a quantity-of-interest functional

$$I_p(f) := \varphi_p(f),$$

where f is a test function in the finite algebra $\mathcal{A}_n$. This is the terminology used throughout the rest of the report: we estimate functionals of a finite probability state, not Born-rule expectation values of self-adjoint quantum operators.

The associated quantum system has Hilbert space $\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}$ and matrix algebra $\mathcal{B}(\mathcal{H}_n)$. A Grover–Rudolph state-preparation circuit A_p implements the pure state

$$A_p|0^n\rangle = \sum_{i=0}^{N-1} \sqrt{p_i} |i\rangle.$$

If $\Pi_i = |i\rangle\langle i|$ is the computational-basis projection, the quantum state $\omega_p(X) = \langle 0^n | A_p^\dagger X A_p | 0^n \rangle$ satisfies

$$\omega_p(\Pi_i) = p_i.$$

This identity is the bridge between the classical probability law on $\mathcal{A}_n$ and the noncommutative quantum experiment on $\mathcal{B}(\mathcal{H}_n)$. Direct distribution experiments estimate the whole vector p . Quantity-of-interest experiments estimate

$$I_p(h) = \sum_{i=0}^{N-1} p_i h(x_i)$$

from measured frequencies $\hat{p}_i$. Event-amplitude experiments choose an event $E \subset D_n$, for example the upper-half event

$$E_{\text{upper}} = \{x_i \in D_n : x_i \geq 1/2\}, \quad a = p(E_{\text{upper}}) = \sum_{x_i \in E_{\text{upper}}} p_i,$$

and then use QAE or MLAE to estimate the scalar amplitude a . For a Grover amplification level k , the ideal marked probability is

$$p_k(a) = \sin^2((2k+1) \arcsin \sqrt{a}).$$

2.5 Grover–Rudolph case names

The name `gr_<profile>_n<n>` has a precise meaning throughout the software and this report. It denotes a Grover–Rudolph probability law on the midpoint grid D_n . A nonnegative profile $F : D_n \rightarrow \mathbb{R}_{\geq 0}$ is sampled on the grid, and the probability vector is

$$p_i = \frac{F(x_i)}{\sum_{j=0}^{2^n-1} F(x_j)}, \quad i = 0, \dots, 2^n - 1.$$

The circuit then prepares $\sum_i \sqrt{p_i} |i\rangle$. For example, `gr_sin2_n3` means $n = 3$, $N = 8$, and $F(x) = \sin^2(\pi x)$. Similarly, `gr_affine_n4` means $n = 4$, $N = 16$, and

$$F(x) = 0.11 + 0.57x, \quad p_i = \frac{0.11 + 0.57x_i}{\sum_{j=0}^{15} (0.11 + 0.57x_j)}.$$

For this affine case, the upper-half event used in the first amplified Grover–Rudolph test has

$$a = \sum_{i=8}^{15} p_i = 0.6803797468 \dots$$

The structural complexity of the angle encoding is tracked through:

- multilinear support, the number of nonzero angle coefficients;
- maximum degree, the largest interaction order in the multilinear angle expansion;
- the minimum m_1 stratum, which records how localized the smallest support term is on the grid;
- the weighted canonical length L_{can} , used as a compact proxy for angle-encoding complexity.

These structural metrics follow the angle-structure viewpoint developed for quantum numerical integration [1]. They complement backend-dependent compilation metrics such as depth, two-qubit depth, and two-qubit gate count. The Phase 1 results show that compilation choices can dominate the hardware behavior even when the logical circuit and the saved transpiled QASM are both ideal-correct.

Finally, the experiment is also informed by two broader theoretical references. The intrinsic gate-embedding perspective of [2] motivates tracking compilation as a mathematical object rather than a purely software detail. The algebraic probability viewpoint of [4] gives the finite-dimensional language in which classical measurement laws, quantum states, and noncommutative operations can be discussed within one probabilistic framework.

3 Methodology

The experimental programme is organized around three complementary workflows. The chronological phase numbering is retained for reproducibility, but the scientific interpretation of the experiments is workflow-based:

Workflow A.

Angle-structured QAE integration. A function $g : [0, 1] \rightarrow [0, 1]$ is encoded as an amplitude under a uniform quadrature rule. The target is the numerical integral $I[g]$, approximated by left, midpoint, right, or Simpson quadrature rules and then estimated through QAE-type circuits.

Workflow B.

Grover–Rudolph probability-law preparation. A positive profile F defines a finite probability law $p_i = F(x_i)/\sum_j F(x_j)$ on a dyadic grid. The hardware prepares $\sum_i \sqrt{p_i}|i\rangle$, measures the resulting law, and computes finite-law quantities of interest $I_p(f) = \sum_i f(x_i)p_i$.

Workflow C.

Backend-aware candidate selection. Circuits from Workflows A and B are admitted to QPU execution only after local construction, transpilation audit, ideal simulation, backend-noise simulation, layout inspection, mitigation selection, and explicit rejection tests.

This separation is essential. Workflow A is the direct continuation of the angle-structured numerical-integration model of [1]. Workflow B is a different experimental object: it validates Grover–Rudolph preparation of a finite probability law and then uses that law for direct algebraic integration. Workflow C is not a separate mathematical estimator; it is the control layer that decides which circuits are credible enough to spend IBM Quantum credits.

3.1 Workflow A: angle-structured QAE integration

For Workflow A, the selected quadrature rule $r \in \{L, M, R\}$ fixes the grid points x_i^r introduced in Section 2.1. The amplitude circuit A_g^r prepares

$$A_g^r|0\rangle^{\otimes n}|0\rangle = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle \left(\sqrt{1 - g(x_i^r)}|0\rangle + \sqrt{g(x_i^r)}|1\rangle \right), \quad N = 2^n.$$

Measuring the final qubit gives

$$P(1; A_g^r) = Q_n^r[g] = \frac{1}{N} \sum_{i=0}^{N-1} g(x_i^r),$$

which is a quadrature estimate of the integral of g . Simpson quadrature is treated as the classical weighted combination of the corresponding left, midpoint, and right estimates, not as a separate support set D_n . The current calibration family g_0, g_1, g_2 belongs to the first angle-structure classes studied in [1]. The planned quadrature extension will separate the discretization error of the rule from the statistical and hardware error of the QAE estimator.

3.2 Workflow B: Grover–Rudolph finite-law integration

For Workflow B, the function used to build the quantum state is not directly the test function to be integrated. A positive profile F defines the finite law

$$p_j^{(n)} = \frac{F(x_j)}{\sum_{\ell=0}^{2^n-1} F(x_\ell)}, \quad x_j = \frac{j + 1/2}{2^n},$$

and the Grover–Rudolph circuit prepares

$$|\psi_p\rangle = \sum_{j=0}^{2^n-1} \sqrt{p_j^{(n)}} |j\rangle.$$

Once the law has been measured, any classical test function f on the same grid defines the finite algebraic quantity

$$I_{p^{(n)}}(f) = \sum_{j=0}^{2^n-1} f(x_j) p_j^{(n)}.$$

This is the interpretation used in the distribution, finite-law integration, MC/QMC baseline, scaling, and Gaussian-profile experiments.

3.3 Workflow C: backend-aware admission rules

Workflow C controls when hardware execution is allowed. It records logical and transpiled circuits, backend identity, physical layout, depth, two-qubit-gate count, two-qubit depth, shots, seeds, expected ideal quantities, ideal simulation, and backend-noise simulation. A candidate is rejected before QPU execution when its depth, two-qubit count, noisy total-variation distance, or largest predicted quantity-of-interest error exceeds the current phase thresholds. The same workflow also records mitigation choices such as dynamical decoupling and randomized compiling/twirling.

3.4 Phase 0: Reproducible audit before spending credits

Phase 0 generates the calibration circuits parametrically from a single Qiskit implementation. The circuit generator follows the dyadic-controlled-rotation logic of Grover–Rudolph state preparation [3], specialized here to the small angle-polynomial test family motivated by [1]. For each case g_i and amplification level k , the audit stores:

- the logical circuit and, when a backend is selected, the transpiled circuit;
- backend name and status;
- logical depth, transpiled depth, two-qubit gate count, two-qubit depth, and operation counts;
- structural metrics of the encoding;
- shot count, seed, and expected ideal probability $p_k(a)$;
- ideal simulation and backend-noise simulation when the corresponding simulator is available.

The audit rejects candidates whose transpiled depth or two-qubit-gate count exceeds predefined thresholds. In this first campaign all g_0, g_1, g_2 cases with $k = 0, 1, 2$ were below the risk thresholds on `ibm_fez`.

3.5 Larger Grover–Rudolph state preparation

The next audit extends the workflow from the three-qubit calibration circuits to genuine Grover–Rudolph loading circuits on larger dyadic grids. Following the finite probability model in Section 2.4, each benchmark starts from a profile $F(x_i)$ on the midpoint grid D_n , normalizes it to a probability vector p_i , and encodes it as

$$\sum_{j=0}^{2^n-1} \sqrt{p_j} |j\rangle.$$

Following the probability-tree construction of [3], for each binary prefix b the generator computes the two child masses $P(b0)$ and $P(b1)$, then applies a conditional rotation satisfying

$$\sin^2(\theta_b/2) = \frac{P(b1)}{P(b0) + P(b1)}.$$

The implementation groups all rotations at a fixed tree level into one uniformly controlled R_y stage. In Qiskit this is represented with `UCRYGate`, matching the ancilla-free Gray-code/Walsh–Hadamard ladder viewpoint of [3]. This is materially different from compiling every prefix as an independent multi-controlled rotation: on the first $n = 4$ audit it reduced the transpiled two-qubit count from hundreds of gates to 16.

3.6 Integration scaling from finite probability laws

After the direct-distribution and cross-backend validation steps, the next methodological question is not whether a deeper Grover iterate can be run, but whether algebraic integration remains stable as the dyadic grid is refined. For a positive profile F on $[0, 1]$, Phase 9 defines the midpoint grid

$$x_j = \frac{j + 1/2}{2^n}, \quad j = 0, \dots, 2^n - 1,$$

and the normalized finite probability law

$$p_j^{(n)} = \frac{F(x_j)}{\sum_{\ell=0}^{2^n-1} F(x_\ell)}.$$

For each classical test function f , the finite algebraic integral is

$$I_{p^{(n)}}(f) = \sum_{j=0}^{2^n-1} f(x_j) p_j^{(n)}.$$

This is compared with the continuous reference

$$I_F(f) = \frac{\int_0^1 f(x) F(x) dx}{\int_0^1 F(x) dx}.$$

The audit therefore separates three effects:

$$\begin{aligned} \epsilon_{\text{disc}}(f) &= \left| I_{p^{(n)}}(f) - I_F(f) \right|, \\ \epsilon_{\text{load}}(f) &= \left| I_{\tilde{p}_{\text{exact}}}(f) - I_{p^{(n)}}(f) \right|, \\ \epsilon_{\text{noise}}(f) &= \left| I_{\tilde{p}_{\text{noise}}}(f) - I_{p^{(n)}}(f) \right|. \end{aligned}$$

Here $\tilde{p}_{\text{exact}}$ is the probability law extracted from the exact logical statevector, and $\tilde{p}_{\text{noise}}$ is the backend-noise simulation after transpilation. A hardware campaign is considered only when the circuit-depth, two-qubit-gate, noisy-TVD, and maximum quantity-of-interest error thresholds are all satisfied.

3.7 Sampling baselines under the same finite law

Phase 10 compares the hardware sampling workflow with classical MC and QMC sampling after the finite law $p^{(n)}$ has been fixed. This is a deliberately controlled comparison: all estimators use the same dyadic law, the same grid, and the same sample budget M . The QPU estimator is

$$I_{\hat{p}_{\text{QPU}}}(f) = \sum_{j=0}^{2^n-1} f(x_j) \hat{p}_{\text{QPU},j},$$

where $\hat{p}_{\text{QPU}}$ is the measured hardware frequency law. The MC baseline draws M independent samples $J_1, \dots, J_M \sim p^{(n)}$ and uses

$$\hat{I}_{\text{MC}}(f) = \frac{1}{M} \sum_{m=1}^M f(x_{J_m}).$$

The QMC baseline uses randomized shifted van-der-Corput points

$$u_m = (v_m + \Delta) \bmod 1, \quad \Delta \sim U(0, 1),$$

and maps them through the inverse CDF of $p^{(n)}$:

$$J_m = \min \left\{ j : \sum_{\ell=0}^j p_{\ell}^{(n)} \geq u_m \right\}, \quad \hat{I}_{\text{QMC}}(f) = \frac{1}{M} \sum_{m=1}^M f(x_{J_m}).$$

This QMC baseline is intentionally strong because it assumes the exact finite CDF is classically available. It should therefore be read as a reference for sampling quality once the law is known, not as a complete cost model for constructing that law. The comparison reports distribution-level errors (TVD, Hellinger distance, classical fidelity, L^2 distance, and maximum cell deviation) and quantity-level MC/QMC RMSE against the absolute QPU error.

3.8 Backend selection

The visible Heron backends were listed before hardware submission. Among the initial candidates, `ibm_fez` had the lowest average two-qubit and readout errors while maintaining zero or low queue depth at the time of the campaign. The first hardware phase was therefore run on one backend only, to avoid confounding amplification-level comparisons with backend calibration differences.

3.9 Compilation as an experimental variable

The $g_0, k = 2$ anomaly observed in Phase 1 shows that compilation cannot be treated as a passive preprocessing step. The logical circuit and the saved QASM were both ideal-correct, but the hardware result changed substantially when the optimization level was changed. This is consistent with the gate-synthesis viewpoint of [2]: elementary operations, their embeddings, and their discretized native-basis realizations are part of the experimental object being benchmarked. For this reason the accepted Phase 1 evidence reports optimization level, depth, two-qubit-gate count, and QASM paths as first-class metadata.

3.10 Hardware execution and stopping rules

Hardware execution is intentionally staged. The first submission used only six circuits: g_0 with $K = \{0, 2\}$, g_1 with $K = \{0, 1\}$, and g_2 with $K = \{0, 1\}$. Deferred circuits were submitted only after the previous results had been fetched and analyzed.

Each hardware run stores a `submitted_campaign.json` file with job identifiers, compilation metrics, expected probabilities, and QASM paths. Result fetching stores raw counts and computes the ancilla success probability $\hat{p}_k$. A compact analysis table then reports the absolute deviation from the ideal probability and a binomial z -score. The classification used in this report is `pass` for $|z| < 4$, `watch` for $4 \leq |z| < 8$ or moderate absolute bias, and `fail` for $|z| \geq 8$ or large absolute bias.

MLAE estimates are computed by maximizing the likelihood of the observed binomial counts across the selected schedule K .

4 Experiments

The experiments are reported in chronological phases for reproducibility, but they should be read through the three-workflow structure introduced in Section 3. Table 1 gives the map used throughout the remainder of the report.

Workflow	Experimental role	Main phases
A	Angle-structured QAE integration of g	Phases 1, 1b, 12–13, planned Phase 16
B	Grover–Rudolph finite-law preparation and direct $I_p(f)$	Phases 3–7, 9–11, 14–15
C	Backend-aware selection, mitigation, and rejection	Phases 0, 2, 6, 8, 11, 13–15

Table 1: Conceptual organization of the experimental phases. The phase numbers remain chronological execution labels; the workflows define the scientific interpretation of each experiment.

4.1 Calibration functions

The first hardware campaign belongs to Workflow A. It uses three two-grid-qubit calibration instances. The ancilla qubit encodes the integrand value $g(x_i)$ through a controlled rotation state preparation, so the measured success probability is a uniform-grid quadrature estimate of g .

Case	Description	a	Degree	Support	L_{can}
g0	constant quarter	0.25	0	1	1.481
g1	affine $\sin^2(\pi x/2)$	0.50	1	3	2.912
g2	quadratic $\sin^2(\pi x)$	0.50	2	4	6.474

Table 2: Calibration functions and structural encoding metrics.

4.2 Phase 1 campaign sequence

All accepted hardware results used `ibm_fez`, 2048 shots per circuit, and seed 12345. The campaign sequence was:

1. an initial six-circuit calibration campaign at optimization level 0;
2. a controlled g_0 repeat at optimization level 2 after detecting a hardware anomaly in $g_0, k = 2$;
3. a full g_1 stress campaign with $K = \{0, 1, 2\}$ at optimization level 2;
4. a full g_2 stress campaign with $K = \{0, 1, 2\}$ at optimization level 2.

Campaign	Schedule	Opt. level	Max depth	Max 2q gates
g0 repeat	$K = \{0, 2\}$	2	71	19
g1 stress	$K = \{0, 1, 2\}$	2	152	36
g2 stress	$K = \{0, 1, 2\}$	2	291	79

Table 3: Accepted Phase 1 hardware campaigns after compilation tuning.

4.3 Grover–Rudolph distribution campaign

After the Phase 1b simulator comparison, the project moves from Workflow A to Workflow B: direct validation of larger Grover–Rudolph state-preparation circuits. This campaign deliberately measures only the prepared distribution: no Grover amplification and no QAE estimator

is applied yet. The purpose is to test whether the hardware reproduces the dyadic probability law before using that law to compute finite-grid quantities $I_p(f)$, or before spending credits on amplified circuits.

The case names use the convention defined in Section 2.5: `gr_<profile>_n<n>` denotes a probability law on 2^n midpoint cells, with probabilities obtained by normalizing a positive profile F on the grid. The exact probability of a cell is therefore not the raw value $F(x_i)$, but

$$p_i = \frac{F(x_i)}{\sum_j F(x_j)}.$$

The concrete profiles used in the larger-grid audit are listed in Table 4. This table is part of the mathematical definition of the experiments.

Case	n	Grid size	Profile $F(x_i)$	Angle class
<code>gr_sin2_half_n3</code>	3	8	$\sin^2(\pi x_i/2)$	$\mathcal{G}_3^{(1)}$
<code>gr_sin2_n3</code>	3	8	$\sin^2(\pi x_i)$	$\mathcal{G}_3^{(2)}$
<code>gr_affine_n4</code>	4	16	$0.11 + 0.57x_i$	$\mathcal{G}_4^{(4)}$
<code>gr_beta_bump_n4</code>	4	16	$x_i(1 - x_i)^4$	$\mathcal{G}_4^{(4)}$

Table 4: Self-contained mathematical definition of the Grover–Rudolph larger-grid cases. The grid points are $x_i = (i + 1/2)/2^n$, and every row uses $p_i = F(x_i)/\sum_j F(x_j)$. The last column reports the angle-structure class obtained when the same sampled profile is interpreted as a QAE amplitude function g , not as a normalized probability law.

This last column is important. The profile $F(x) = 0.11 + 0.57x$ is affine as a classical function of x , but the associated QAE angle map $2 \arcsin \sqrt{F(x_i)}$ has full multilinear degree on the $n = 4$ grid. Thus “affine profile” and “affine angle encoding” are different statements. The hardware interpretation of the larger Grover–Rudolph experiments is therefore not that all later cases lie in the first classes of [1]; rather, the experiments deliberately compare the low-degree calibration hierarchy g_0, g_1, g_2 with richer finite probability laws whose direct state-preparation circuits are still short because of the Grover–Rudolph probability-tree structure.

The prepared campaign uses `ibm_fez`, 2048 shots per circuit, optimization level 2, seed 12345, and the UCRY implementation of the Grover–Rudolph stages. The three selected circuits are listed in Table 5.

Case	Profile	Qubits	Grid	Depth	2q gates
<code>gr_sin2_n3</code>	$\sin^2(\pi x)$	3	8	31	7
<code>gr_sin2_half_n3</code>	$\sin^2(\pi x/2)$	3	8	36	7
<code>gr_affine_n4</code>	$0.11 + 0.57x$	4	16	76	16

Table 5: Prepared Grover–Rudolph distribution-measurement campaign on `ibm_fez`. The campaign measures the output distribution only.

The deferred case is `gr_beta_bump_n4`. It is viable by depth and two-qubit count, but its localized probability profile made it slightly more sensitive in the noisy Phase 0-GR audit.

4.4 Minimal amplified Grover–Rudolph test

After the mitigated distribution repeat, the first amplified test is defined as a minimal MLAE campaign on `gr_affine_n4`. The marked event is the upper half of the dyadic grid,

$$E_{\text{upper}} = \{x_i = (i + 1/2)/16 : i = 8, \dots, 15\}.$$

For the affine probability law

$$p_i = \frac{0.11 + 0.57x_i}{\sum_{j=0}^{15} (0.11 + 0.57x_j)},$$

this event has exact probability

$$a = \sum_{i=8}^{15} p_i = 0.6803797468 \dots$$

The first schedule is $K = \{0, 1\}$; $k = 2$ remains deferred until these two circuits are retrieved.

Case	k	Expected p_k	Depth	2q gates	2q depth
gr_affine_n4	0	0.680380	74	16	16
gr_affine_n4	1	0.052765	225	51	51

Table 6: Prepared minimal amplified Grover–Rudolph MLAЕ campaign on `ibm_fez`. Runtime options match the mitigated distribution repeat.

4.5 Phase 9 integration-scaling audit

The Phase 9 audit is credit-free. It studies whether direct Grover–Rudolph probability loading can support algebraic integration as the grid is refined from 2^3 to 2^5 cells. The initial cases are listed in Table 7. The affine $n = 4$ law is included as the cross-backend anchor from Phases 7 and 8; the $n = 5$ laws are not submitted to hardware unless the backend-aware audit predicts acceptable cost and quantity-of-interest error.

Case	Profile $F(x)$	Grid	Angle class	Role
gr_sin2_n3	$\sin^2(\pi x)$	8	$\mathcal{G}_3^{(2)}$	baseline
gr_sin2_n4	$\sin^2(\pi x)$	16	$\mathcal{G}_4^{(2)}$	refinement
gr_sin2_n5	$\sin^2(\pi x)$	32	$\mathcal{G}_5^{(2)}$	stress
gr_affine_n4	$0.11 + 0.57x$	16	$\mathcal{G}_4^{(4)}$	cross-backend anchor
gr_affine_n5	$0.11 + 0.57x$	32	$\mathcal{G}_5^{(5)}$	refinement
gr_beta_bump_n4	$x(1-x)^4$	16	$\mathcal{G}_4^{(4)}$	localized diagnostic
gr_beta_bump_n5	$x(1-x)^4$	32	$\mathcal{G}_5^{(5)}$	localized stress

Table 7: Initial Phase 9 grid-scaling cases for direct algebraic integration. Each profile is normalized on the dyadic midpoint grid before state preparation. The angle class again refers to the raw sampled profile interpreted as a QAE amplitude function.

The tested quantities of interest are the upper-half mass, $I_p(x)$, $I_p(x^2)$, $I_p(\sin(\pi x))$, and the call payoff $I_p((x - 1/2)_+)$. The audit writes both a case-level candidate table and a quantity-level error table. A row is promoted to a hardware candidate only when it satisfies all structural and noise-model thresholds; amplified QAE remains outside this phase.

4.6 Phase 10 sampling-baseline comparison

Phase 10 uses no additional QPU credits. It reuses the fetched Phase 9 counts for the 32-cell affine law and the 32-cell beta-like stress law on `ibm_kingston`. For each case, the hardware sample size is $M = 8192$. The comparison then generates 1000 independent MC trials and 1000

randomized QMC trials with the same M , always sampling from the exact finite law $p^{(n)}$. The tested quantities are again the upper-half mass, $I_p(x)$, $I_p(x^2)$, $I_p(\sin(\pi x))$, and $I_p((x - 1/2)_+)$.

The purpose is not to claim a quantum advantage. The purpose is to locate the hardware estimates relative to the ordinary finite-law sampling scale. The MC baseline represents direct random sampling from $p^{(n)}$. The QMC baseline uses the inverse CDF of $p^{(n)}$, and is therefore a strong reference once the finite law is known. A QPU estimate is considered competitive with MC for a given test function only when its absolute error is comparable with the MC RMSE at the same sample count.

4.7 Phase 11 dual angle-structure strategy

Phase 11 is again credit-free. It converts the discussion of Section 2.3 into a decision table for pairs (F, f) , where F defines the Grover–Rudolph probability law and f is the test function in $I_p(f)$. For each pair the script records:

- the angle class of the sampled profile F , interpreted as a QAE amplitude function;
- the angle class of the test function f , also interpreted as a QAE amplitude function;
- the Phase 9 direct-sampling audit status, when available;
- the Phase 10 MC/QMC comparison status, when available;
- a conservative recommendation: direct sampling, simulator-only QAE, amplified-QAE candidate, diagnostic, or deferred QPU repeat.

The rule is intentionally asymmetric. Low angle degree can promote a pair to a simulator QAE candidate, but poor Phase 10 evidence can still veto additional QPU spending. Conversely, high angle degree does not prevent direct Grover–Rudolph integration, because direct sampling estimates $I_p(f)$ from the measured finite law rather than loading f as an amplitude oracle.

4.8 Phase 12 local QAE-amplitude audit

Phase 12 checks the Phase 11 amplified-QAE candidates by constructing the actual amplitude circuit before any backend audit. Given a finite law p and a test function $f \in [0, 1]$, the preparation block maps

$$|0\rangle \mapsto \sum_i \sqrt{p_i} |i\rangle \left(\sqrt{1 - f(x_i)} |0\rangle + \sqrt{f(x_i)} |1\rangle \right).$$

The objective-qubit probability is therefore $I_p(f)$. For each selected pair the audit builds $Q^k A|0\rangle$, $k = 0, 1, 2$, verifies the exact statevector probability against

$$p_k(a) = \sin^2((2k + 1) \arcsin \sqrt{a}), \quad a = I_p(f),$$

and transpiles locally to the **cz**, **rz**, **sx**, and **x** basis. The audit also flags degenerate schedules where $p_k(a)$ is constant over the selected k values, as happens for $a = 1/2$.

4.9 Phase 13 backend-aware amplified-QAE veto

Phase 13 keeps the same no-credit policy but adds backend-derived noisy simulation for the only non-degenerate Phase 12 amplified-QAE candidate: **gr_sin2_n3** with $f(x) = \sin(\pi x)$. The script constructs the augmented finite-law amplitude circuit, transpiles it against the selected IBM backend, simulates with the corresponding noise model, and accepts a schedule only if all required k -levels pass the depth, two-qubit-gate, and noisy probability thresholds. No QPU jobs are submitted in this phase.

4.10 Phase 14 direct-sampling scaling audit

Phase 14 returns to direct Grover–Rudolph integration, because this was the hardware-facing workflow retained after Phases 10–13. The audit is again credit-free and compares $n = 4$, $n = 5$, and $n = 6$ midpoint grids for three probability-profile families: the smooth symmetric profile $\sin^2(\pi x)$, the positive affine profile

$$F_{\text{aff}}(x) = 0.11 + 0.57x,$$

and the localized beta-like stress profile

$$F_{\text{beta}}(x) = x(1 - x)^4.$$

For each profile the script builds the Grover–Rudolph state-preparation circuit, transpiles locally to the `cz`, `id`, `rz`, `sx`, and `x` basis over several optimization levels and seeds, and records the best two-qubit-gate/depth candidate. It also computes finite-law values $I_p(f)$, continuous midpoint-reference errors, MC and randomized-QMC sampling baselines at the same shot count, and the angle class of the profile and each tested quantity of interest. A row promoted by this phase is only a candidate for a later backend-aware audit, not for immediate QPU submission.

4.11 Phase 15 alternative smooth probability-law screening

Phase 15 closes the direct finite-law integration block by asking whether the observed affine $n = 6$ bias is specific to the one-sided affine probability law or persists for smoother centered laws. The tested reference candidate is a Gaussian-like profile on the 64-point midpoint grid,

$$F_{\text{gauss}}(x) = \exp\left(-\frac{1}{2} \left(\frac{x - 1/2}{0.25}\right)^2\right), \quad p_i = \frac{F_{\text{gauss}}(x_i)}{\sum_j F_{\text{gauss}}(x_j)}.$$

As in Phase 14, the audit first performs local and backend-aware transpilation and noisy simulation. A QPU campaign is submitted only after the candidate passes the backend-aware screening. The measured distribution is then evaluated through the same five quantities of interest:

$$1_{x \geq 1/2}, \quad x, \quad x^2, \quad \sin(\pi x), \quad (x - 1/2)_+.$$

4.12 Transition to quadrature-rule QAE integration

Phases 3–15 study direct Grover–Rudolph finite-law integration:

$$I_p(f) = \sum_i p_i f(x_i),$$

where p is a measured probability law. This is a useful hardware workflow, but it is distinct from the numerical-integration protocol analysed in the angle-structure paper. In that protocol the integrand g itself is encoded as an amplitude and the quadrature rule is explicit: left Riemann, midpoint, right Riemann, or Simpson. The next experimental block therefore returns to

$$I[g] = \int_0^1 g(x) dx, \quad Q_n^{(p)}[g],$$

and separates four terms: quadrature rule, discretization error, QAE/MLAE estimation error, and encoding-circuit cost.

4.13 Phase 15b quadrature-rule QAE microcampaign

The first hardware test of this returned Workflow-A formulation used midpoint quadrature only. The goal was not yet to compare all quadrature rules on hardware, but to verify that the new scripts correctly separate

$$Q_n[g], \quad I[g], \quad \hat{Q}_n[g] - Q_n[g], \quad \hat{Q}_n[g] - I[g].$$

The selected cases are listed in Table 8. They were chosen to include the two non-degenerate calibration functions g_1, g_2 , one non-special affine amplitude, and one smooth low-amplitude stress function.

Case	Rule	n	Schedule K	Max depth	Max 2q gates
g1	midpoint	2	$\{0, 1\}$	72	19
g2	midpoint	2	$\{0, 1, 2\}$	99	32
affine_shifted	midpoint	2	$\{0, 1\}$	83	20
smooth_bump	midpoint	3	$\{0, 1, 2\}$	304	76

Table 8: Phase 15b quadrature-rule QAE microcampaign on `ibm_kingston`. The circuits used 2048 shots per amplification level, optimization level 2, dynamical decoupling, and randomized compiling/twirling.

The campaign is defined by

`experiments/phase15b_quadrature_microcampaign_ibm_kingston.json`

and executed through

`scripts/submit_quadrature_qae_campaign.py`.

The result fetcher

`scripts/fetch_quadrature_qae_results.py`

groups the hardware probabilities by case and amplification schedule, performs the MLAE likelihood inversion, and stores both circuit-level probabilities and case-level quadrature estimates.

5 Results

The results are grouped chronologically because each hardware decision depends on the previous audit, but the interpretation follows the three workflows: Workflow A tests angle-structured QAE integration of g , Workflow B tests Grover–Rudolph finite-law preparation and direct evaluation of $I_p(f)$, and Workflow C explains why some circuits were accepted, repeated, mitigated, or rejected before further QPU execution.

5.1 Initial anomaly and diagnosis

The first `ibm_fez` campaign at optimization level 0 revealed a strong Workflow-A/Workflow-C interaction in $g_0, k = 2$. The observed hardware probability was $\hat{p} = 0.597656$, while the expected value is $p = 0.25$. With 2048 shots this corresponds to approximately 36 binomial standard deviations, and therefore cannot be treated as shot noise.

The diagnostic script verified three facts:

- the logical $g_0, k = 2$ circuit has exact ideal probability 0.25;

- the saved transpiled QASM from the failed run also has exact ideal probability 0.25;
- backend-noise simulation predicted a moderate bias near 0.28, not the observed 0.60.

The failure was therefore classified as a compilation-sensitive hardware effect, not a logical circuit or QASM convention error.

5.2 Accepted hardware circuits

Table 9 reports the accepted Workflow-A calibration circuits after applying the Workflow-C admission and repeat rules.

Case	k	Depth	2q gates	p	$\hat{p}$	Abs. dev.	Status
g0	0	5	0	0.250000	0.246094	0.003906	pass
g0	2	71	19	0.250000	0.297363	0.047363	watch
g1	0	18	4	0.500000	0.504395	0.004395	pass
g1	1	78	18	0.500000	0.530762	0.030762	pass
g1	2	152	36	0.500000	0.511230	0.011230	pass
g2	0	43	9	0.500000	0.506836	0.006836	pass
g2	1	167	43	0.500000	0.516113	0.016113	pass
g2	2	291	79	0.500000	0.504883	0.004883	pass

Table 9: Accepted Phase 1 hardware circuit results on `ibm_fez`.

The controlled g_0 repeat reduced the $g_0, k = 2$ anomaly from $\hat{p} = 0.597656$ to $\hat{p} = 0.297363$ by changing from optimization level 0 to optimization level 2. The repeated point remains in the watch region, but the corresponding MLAE estimate becomes usable.

5.3 MLAE estimates

Case	Schedule	Exact a	Hardware $\hat{a}$	Abs. error
g0	$K = \{0, 2\}$	0.250000	0.241051	0.008949
g1	$K = \{0, 1, 2\}$	0.500000	0.499093	0.000907
g2	$K = \{0, 1, 2\}$	0.500000	0.499512	0.000488

Table 10: Accepted Phase 1 MLAE estimates.

The affine and quadratic stress tests are the strongest results of Phase 1. Both pass through $k = 2$ at optimization level 2, and both produce sub- 10^{-3} MLAE absolute errors. The g_2 campaign is especially important because it is the structurally harder case, with higher degree, larger canonical length, and the deepest accepted circuit.

5.4 Reproducibility artifacts

The accepted Phase 1 summary is generated by:

```
python3 scripts/summarize_phase1_results.py.
```

The corresponding CSV and JSON artifacts are stored under:

```
results/hardware/phase1_summary/.
```

5.5 Phase 1b simulator comparison

The first Phase 1b statistical simulator belongs to Workflow A. It compares MLAE, an IQAE-style adaptive likelihood proxy, canonical grid QAE, Monte Carlo, and quasi-Monte Carlo on the same g_0, g_1, g_2 calibration family. For the fixed-schedule hardware candidate, $K = \{0, 1, 2\}$, the simulator gives mean absolute errors near 10^{-3} , consistent with the hardware regime observed for g_1 and g_2 .

Case	Method	Variant	MAE	RMSE	Queries
g0	IQAE proxy	steps=3	0.000764	0.000950	30597
g0	MLAE	$K = \{0, 1, 2\}$	0.001238	0.001556	18432
g1	IQAE proxy	steps=3	0.000634	0.000802	38912
g1	MLAE	$K = \{0, 1, 2\}$	0.001511	0.001931	18432
g2	IQAE proxy	steps=3	0.000748	0.000935	38912
g2	MLAE	$K = \{0, 1, 2\}$	0.001459	0.001815	18432

Table 11: Phase 1b simulator comparison between MLAE and an IQAE-style adaptive likelihood proxy. Query counts are oracle-query proxies under the ideal binomial model.

The adaptive proxy improves ideal statistical error but uses a larger oracle-query budget. Canonical QAE and MC/QMC are not decisive on this small calibration family: the amplitudes $a = 0.5$ and the four-point grid create exact or near-exact degeneracies for several variants. The next simulator comparison should therefore include non-special amplitudes or larger structured Grover–Rudolph instances.

The non-degenerate Phase 1b extensions confirm this conclusion. Synthetic amplitudes $a = 0.17, 0.33, 0.61$ and larger midpoint-grid functions remove the most obvious special-amplitude degeneracies. Representative results are shown in Table 12.

Case	Method	Variant	MAE	Queries
a17	IQAE proxy	steps=3	0.000553	38912
a17	MLAE	$K = \{0, 1, 2\}$	0.001067	18432
a33	IQAE proxy	steps=3	0.000692	38912
a33	MLAE	$K = \{0, 1, 2\}$	0.001357	18432
a61	IQAE proxy	steps=3	0.000703	36823
a61	MLAE	$K = \{0, 1, 2\}$	0.001446	18432
gr_affine_n4	IQAE proxy	steps=3	0.000867	33710
gr_affine_n4	MLAE	$K = \{0, 1, 2\}$	0.001443	18432
gr_beta_bump_n4	IQAE proxy	steps=3	0.000512	38359
gr_beta_bump_n4	MLAE	$K = \{0, 1, 2\}$	0.001035	18432

Table 12: Phase 1b non-degenerate simulator extensions. IQAE-style adaptation improves ideal statistical error but uses a substantially larger oracle-query budget than fixed-schedule MLAE.

5.6 Phase 0-GR audit for larger state preparation

The first larger Grover–Rudolph audit validates the state loader before any QAE amplification is attempted. Exact statevector checks give total variation distance at numerical roundoff level, confirming that the dyadic probability tree and uniformly controlled R_y implementation prepare the intended probability law. The backend-aware audit on `ibm_fez` then gives the cost and noisy-distribution estimates in Table 13.

Case	Qubits	Grid	Depth	2q gates	Noisy TVD
<code>gr_sin2_half_n3</code>	3	8	36	7	0.0241
<code>gr_sin2_n3</code>	3	8	31	7	0.0200
<code>gr_affine_n4</code>	4	16	76	16	0.0281
<code>gr_beta_bump_n4</code>	4	16	70	16	0.0320

Table 13: Phase 0-GR backend-aware audit on `ibm_fez`. All exact statevector total variation distances are below 10^{-15} ; the table reports the noisy Aer readout extracted from the backend model.

All four circuits pass the configured depth and two-qubit thresholds. The first hardware distribution campaign therefore selects `gr_sin2_n3`, `gr_sin2_half_n3`, and `gr_affine_n4`. The decision is deliberately conservative: it tests two low-cost eight-point distributions and one sixteen-point non-symmetric distribution before adding QAE amplification.

5.7 First Grover–Rudolph distribution hardware readout

The first distribution-measurement campaign was run on `ibm_fez` with 2048 shots per circuit. It measures the prepared distribution directly and does not include QAE amplification. The results are summarized in Table 14.

Case	Qubits	Depth	2q gates	Phase 0 noisy TVD	Hardware TVD
<code>gr_sin2_half_n3</code>	3	36	7	0.0241	0.0216
<code>gr_sin2_n3</code>	3	31	7	0.0200	0.0313
<code>gr_affine_n4</code>	4	76	16	0.0281	0.0479

Table 14: First hardware measurement of larger Grover–Rudolph prepared distributions on `ibm_fez`. TVD denotes total variation distance between the measured distribution and the target dyadic probability law.

The two eight-point circuits are stable at the few-percent TVD level. The sixteen-point affine circuit remains viable but shows the expected increase in noise sensitivity, with measured TVD 0.0479 versus a noisy-simulation estimate 0.0281. This motivates a second microcampaign focused only on `gr_affine_n4`, preferably with light Runtime mitigation and no QAE amplification, to distinguish sampling variation from a systematic state-preparation bias.

The focused repeat used 4096 shots with dynamical decoupling and gate/measurement twirling. It reduced the sixteen-point affine TVD from 0.0479 to 0.0341, and the maximum per-cell deviation from 0.0235 to 0.0086. This result is close to the Phase 0-GR noisy estimate and supports using the same Runtime options for the first amplified test.

Run	Shots	Runtime mitigation	TVD	Max. dev.
First <code>gr_affine_n4</code> readout	2048	no	0.0479	0.0235
Focused repeat	4096	DD + twirling	0.0341	0.0086

Table 15: Focused repeat of the first sixteen-point Grover–Rudolph distribution on `ibm_fez`.

5.8 Mitigated distribution suite

After the amplified `gr_affine_n4` diagnosis showed that the first Grover iterate is the fragile block, a second distribution-only suite was run with the same Runtime mitigation used in

the affine repeat. This suite has two roles: it checks whether mitigation is harmless on the already stable eight-point cases, and it gives the first hardware readout of the deferred localized sixteen-point profile `gr_beta_bump_n4`.

Case	Shots	Depth	2q	Raw TVD	Mit. TVD	Mit. max dev.
<code>gr_sin2_half_n3</code>	4096	33	7	0.0216	0.0208	0.0100
<code>gr_sin2_n3</code>	4096	31	7	0.0313	0.0368	0.0200
<code>gr_beta_bump_n4</code>	4096	68	16	–	0.0456	0.0241

Table 16: Phase 5 mitigated Grover–Rudolph distribution suite on `ibm_fez`. “Mit.” denotes the run with dynamical decoupling and twirling. The raw TVD column refers to the first unmitigated distribution campaign when available.

The result is deliberately interpreted as mixed evidence. Mitigation is useful for the sixteen-point affine distribution and neutral for `gr_sin2_half_n3`, but it worsens `gr_sin2_n3` from TVD 0.0313 to 0.0368. Therefore DD and twirling should not be treated as automatic defaults for every short distribution circuit. They are useful experimental variables that must be reported case by case.

The first `gr_beta_bump_n4` hardware run is viable but close to the current risk boundary: TVD is 0.0456 and the largest single-cell deviation is 0.0241. Direct quantities of interest remain informative but functional-dependent. For `gr_beta_bump_n4`, the measured mean x is accurate within 0.0012, while $\sin(\pi x)$ has error 0.0177 and the second moment has error 0.0055. This confirms that localized profiles should be benchmarked through quantity-of-interest error, not only global distribution distance.

A follow-up credit-free audit was then run for the exact localized Phase 5 profile $F(x) = x(1-x)^4$. This distinction matters because the earlier Phase 0-GR beta-like profile used different beta parameters. The audit correctly predicted that the executed candidate was risky: opt-2, seed 12345 gave noisy TVD 0.0439, close to the measured hardware TVD 0.0456. It suggested opt-3, seed 45678 as a better candidate, with the same 16 two-qubit gates and predicted noisy TVD 0.0217. This candidate was submitted as a one-circuit focused repeat.

Run	Depth	2q	TVD	Max. dev.
Phase 5 opt-2, seed 12345	68	16	0.0456	0.0241
Phase 5b opt-3, seed 45678	70	16	0.0502	0.0170

Table 17: Focused `gr_beta_bump_n4` repeat. The repeat reduced the largest single-cell deviation but did not reduce total variation distance.

The focused repeat does not validate the optimistic noisy-simulation prediction: TVD increases slightly from 0.0456 to 0.0502, although the maximum cell deviation decreases from 0.0241 to 0.0170. Errors for quantities of interest are also mixed: $\sin(\pi x)$ improves from 0.0177 to 0.0150, but the mean, second moment, upper-half event, and call payoff all worsen. The operational decision is therefore conservative: localized sixteen-point profiles remain viable for direct distribution diagnostics, but they should not be promoted to amplified QAE on the current backend.

5.9 Phase 6 direct-distribution postprocessing

Phase 6 starts without further QPU execution. Motivated by the mixed-state preparation reference of [5], the analysis treats the Grover–Rudolph loader as an eigenvalue-encoding layer for a diagonal probability law. This motivates adding distribution metrics beyond TVD and testing a simulator-only purification diagnostic.

The extended postprocessing computes classical fidelity, Hellinger distance, Euclidean distance, maximum cell deviation, and bootstrap confidence intervals from the existing hardware counts. Table 18 summarizes the first readout, and Figure 1 visualizes the two main distribution distances. The color convention in this figure is categorical, not quantitative. For each case–variant pair, the left bar is the TVD and the right bar is the Hellinger distance. The height of the bar gives the numerical distance. The color of the TVD bar identifies the experimental variant: raw execution, mitigated execution, or opt-3 repeat. The Hellinger bar is always grey because it is not a different experimental condition; it is a second distance metric computed from the same measured counts. This avoids assigning two unrelated colors to the same hardware run.

Case	Variant	TVD	Hellinger	F_{cl}	L^2
gr_affine_n4	raw	0.0479	0.0470	0.9956	0.0332
gr_affine_n4	mitigated	0.0341	0.0281	0.9984	0.0198
gr_beta_bump_n4	mitigated	0.0456	0.0629	0.9921	0.0330
gr_beta_bump_n4	opt-3 repeat	0.0502	0.0703	0.9901	0.0315
gr_sin2_half_n3	raw	0.0216	0.0221	0.9990	0.0185
gr_sin2_half_n3	mitigated	0.0208	0.0317	0.9980	0.0173
gr_sin2_n3	raw	0.0313	0.0401	0.9968	0.0248
gr_sin2_n3	mitigated	0.0368	0.0458	0.9958	0.0306

Table 18: Phase 6 distribution metrics from existing hardware counts. Here $F_{\text{cl}} = (\sum_i \sqrt{p_i \hat{p}_i})^2$ is the classical fidelity and L^2 is the Euclidean distance between target and measured distributions.

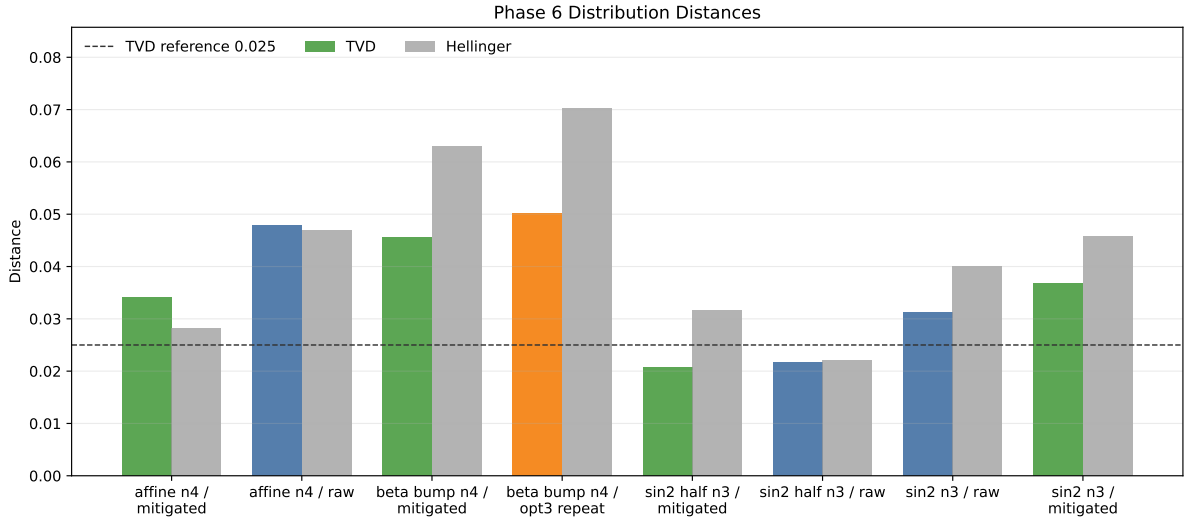


Figure 1: Phase 6 TVD and Hellinger distances for the available Grover–Rudolph distribution measurements. TVD bar colors encode the experimental variant: raw, mitigated, or opt-3 repeat. Hellinger bars are grey to emphasize that they report a second distance metric for the same case–variant pair, not a distinct hardware run.

The additional metrics confirm the Phase 5 interpretation. The mitigated affine case improves across all reported distances. The opt-3 repeat of **gr_beta_bump_n4** lowers the Euclidean distance slightly but worsens TVD, Hellinger distance, and classical fidelity. Therefore it is not a robust improvement. Bootstrap analysis of quantities of interest also confirms that the largest remaining errors are functional-specific, especially upper-half events, moments, and $\sin(\pi x)$ for selected profiles.

The purification diagnostic is simulator-only. It prepares p on a main register, copies the computational-basis index to an auxiliary register with CNOT gates, and measures both registers. In the ideal simulator, the two marginals agree and the mismatch probability is zero. The first diagnostics use 6 qubits for the $n = 3$ profiles and 8 qubits for `gr_beta_bump_n4`. The observed simulator depths are 12, 13, and 28, respectively, with zero mismatch. This provides a clean conceptual bridge between Grover–Rudolph probability loading, algebraic probability, and purification-based mixed-state preparation, without spending hardware credits.

The resulting decision table is shown in Table 19. The classification is deliberately conservative: a profile is promoted only as a direct distribution result unless the amplified Grover iterate has also passed the hardware stability tests.

Case	Variant	TVD	Max QoI err.	Decision
<code>gr_affine_n4</code>	mitigated	0.0341	0.0112	selected direct functionals
<code>gr_affine_n4</code>	raw	0.0479	0.0329	diagnostic only
<code>gr_beta_bump_n4</code>	mitigated	0.0456	0.0177	diagnostic only
<code>gr_beta_bump_n4</code>	opt-3 repeat	0.0502	0.0150	diagnostic only
<code>gr_sin2_half_n3</code>	mitigated	0.0208	0.0186	direct distribution
<code>gr_sin2_half_n3</code>	raw	0.0216	0.0196	direct distribution
<code>gr_sin2_n3</code>	raw	0.0313	0.0098	direct distribution with caveat
<code>gr_sin2_n3</code>	mitigated	0.0368	0.0178	diagnostic only

Table 19: Phase 6 operational decisions from distribution and quantity-of-interest postprocessing. No row is promoted to amplified QAE on the current evidence.

Table 20 gives the strongest direct quantity-of-interest estimates selected across the available raw, mitigated, and repeat variants. The table lists the test function $f \in \mathcal{A}_n$; the reported estimate is the algebraic-probability functional $I_{\hat{p}}(f) = \varphi_{\hat{p}}(f)$ evaluated on the measured frequency law, and the exact value is $I_p(f) = \varphi_p(f)$ evaluated on the target dyadic law. These are the functionals that should be emphasized before any additional hardware spending. Across the twenty selected case–functional pairs, five are classified as publishable, seven as usable with a confidence interval, five as usable only with an explicit bias report, and three as diagnostic. Figure 2 shows the full ordered functional-error profile. Here the color convention changes because the plotted object is different. Each bar is one selected functional $I_{\hat{p}}(f)$, and the height of the bar is the absolute error $|I_{\hat{p}}(f) - I_p(f)|$. The color does not encode the circuit variant; it encodes the decision status assigned after comparing the error and the bootstrap interval with the exact value. Green denotes publishable, blue denotes usable with confidence-interval coverage, orange denotes usable only with explicit bias reporting, purple denotes diagnostic, and red denotes rejected. Thus the alternating colors in Figure 2 reflect the quality class of each functional, not an alternating sequence of methods.

Case	Test function f	$I_{\hat{p}}(f)$	$I_p(f)$	Status
<code>gr_beta_bump_n4</code>	$f(x) = x$	0.28552	0.28434	publishable
<code>gr_sin2_half_n3</code>	$f(x) = (x - 1/2)_+$	0.22211	0.22566	publishable
<code>gr_sin2_half_n3</code>	$f(x) = \sin(\pi x)$	0.64328	0.64073	publishable
<code>gr_sin2_n3</code>	$f(x) = (x - 1/2)_+$	0.07513	0.07571	publishable
<code>gr_sin2_n3</code>	$f(x) = x^2$	0.28152	0.28274	publishable

Table 20: Best Phase 6 direct quantity-of-interest functionals. The selected variant is the one with the smallest absolute error in $|I_{\hat{p}}(f) - I_p(f)|$ among the available hardware runs.

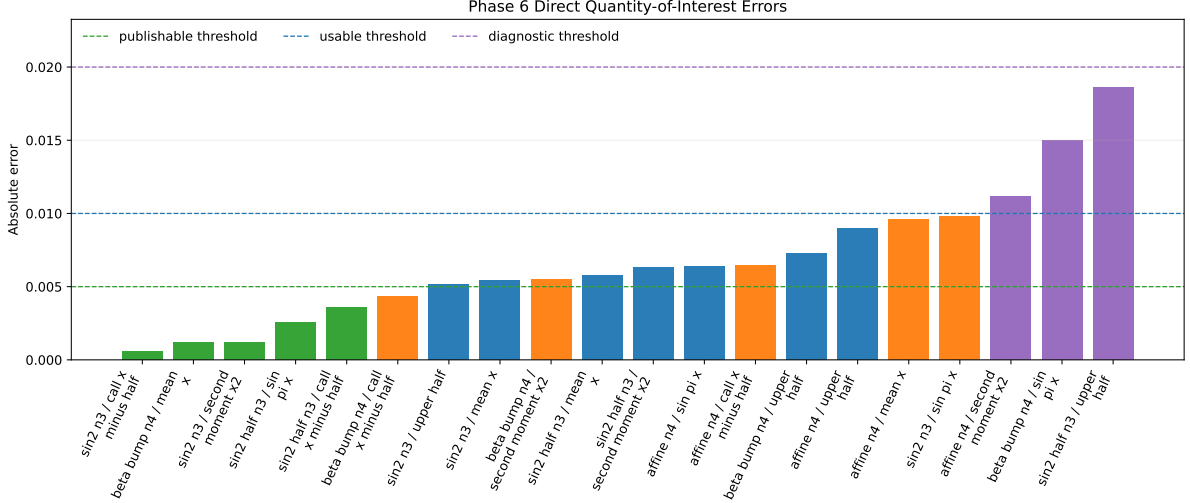


Figure 2: Best direct quantity-of-interest errors selected across raw, mitigated, and repeat variants. Bar colors encode the decision status assigned to each functional: publishable, usable with confidence interval, usable with explicit bias reporting, diagnostic, or rejected. The horizontal lines indicate the publishable, usable, and diagnostic thresholds used by the Phase 6 decision table.

5.10 Higher-shot validation of selected $n = 3$ distributions

Phase 7 repeated the two $n = 3$ direct Grover–Rudolph distribution measurements with 8192 shots and no Runtime mitigation. The purpose was not to improve the global state-preparation claim by averaging alone, but to test run-to-run stability of the finite probability law before escalating to deeper amplified circuits. The result is summarized in Table 21.

Case	Shots	TVD	Hellinger	Fidelity	Max. cell dev.
gr_sin2_half_n3	8192	0.030093	0.035326	0.997506	0.022513
gr_sin2_n3	8192	0.056027	0.056296	0.993672	0.027206
gr_affine_n4 mitigated	8192	0.028017	0.023874	0.998860	0.008197

Table 21: Phase 7 higher-shot validation on **ibm_fez**. The two $n = 3$ runs are raw distribution measurements; the **gr_affine_n4** run uses the same dynamical-decoupling and twirling options as the earlier mitigated campaign. No Grover amplification or QAE circuit is used.

The higher-shot repeat does not reduce the discrepancy relative to the earlier Phase 3 campaign. In particular, **gr_sin2_n3** crosses the predefined TVD threshold of 0.04 and is therefore retained as diagnostic evidence rather than promoted as a global distribution benchmark. The quantity-of-interest analysis is more nuanced: $I_p(x)$ for **gr_sin2_n3** remains publishable with absolute error 0.001556, while $I_p((x - 1/2)_+)$ and $I_p(x^2)$ remain usable with explicit bias reporting. For **gr_sin2_half_n3**, the call payoff and $\sin(\pi x)$ functionals remain below 10^{-2} , but the upper-half event is rejected in this repeat. This reinforces the central methodological conclusion: on current hardware, direct Grover–Rudolph loading should be reported through quantity-specific functionals $I_{\hat{p}}(f)$, with run-to-run stability checks, rather than as a blanket distribution-preparation success.

The mitigated **gr_affine_n4** repeat gives the cleanest Phase 7 global distribution metric: TVD improves from 0.034112 in Phase 6 to 0.028017, and the maximum cell deviation stays below 10^{-2} . The functional picture is again more selective. The $\sin(\pi x)$ functional remains usable with confidence interval coverage and absolute error 0.006274, and the call payoff remains

below 10^{-2} with explicit bias reporting. In contrast, $I_p(x)$, $I_p(x^2)$, and the upper-half event are diagnostic or rejected in this repeat, with the upper-half error increasing to 0.021078. Thus, even when global distribution distances improve, the admissible conclusions remain tied to the specific test function f .

Combining the Phase 3 baseline, Phase 6 postprocessing, and Phase 7 higher-shot repeats gives the paper-quality classification in Table 22. The classification is deliberately conservative: a quantity is paper-quality only if the Phase 7 repeat either preserves confidence-interval coverage or remains below 10^{-2} with an explicit bias caveat.

Category	Result	Decision
Distribution	gr_affine_n4 mitigated	paper-quality distribution
$I_p(f)$	gr_affine_n4, $f(x) = \sin(\pi x)$	paper-quality
$I_p(f)$	gr_sin2_n3, $f(x) = x$	paper-quality
$I_p(f)$	gr_affine_n4, $f(x) = (x - 1/2)_+$	paper-quality with bias caveat
$I_p(f)$	gr_beta_bump_n4, $f(x) = (x - 1/2)_+$	paper-quality with bias caveat
$I_p(f)$	gr_beta_bump_n4, $f(x) = x$	paper-quality with bias caveat
$I_p(f)$	gr_sin2_half_n3, $f(x) = (x - 1/2)_+$	paper-quality with bias caveat
$I_p(f)$	gr_sin2_half_n3, $f(x) = \sin(\pi x)$	paper-quality with bias caveat
$I_p(f)$	gr_sin2_n3, $f(x) = (x - 1/2)_+$	paper-quality with bias caveat
$I_p(f)$	gr_sin2_n3, $f(x) = x^2$	paper-quality with bias caveat

Table 22: Phase 3/6/7 classification of direct Grover–Rudolph hardware evidence. Results not listed here are retained as backend diagnostics under the current thresholds.

Figure 3 shows the final distribution-level decision. The green bar is the only global distribution claim retained for the paper-quality set. Purple bars are still scientifically useful, but only as backend diagnostics: their full measured probability laws are not stable enough for a global state-preparation claim on `ibm_fez`.

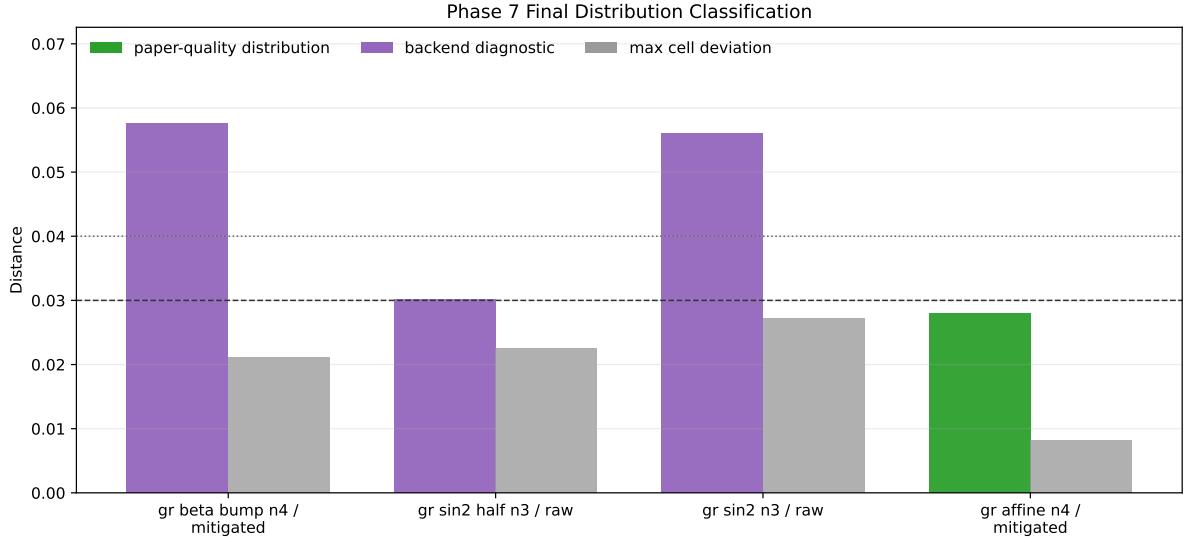


Figure 3: Final Phase 7 distribution-level classification. Bar height reports TVD or maximum cell deviation. Color reports the final distribution decision: paper-quality distribution or backend diagnostic distribution.

Figure 4 gives the corresponding functional-level decision. This is the main paper-facing figure for the direct Grover–Rudolph workflow: it separates quantities that are robust enough

for the main text, quantities that require explicit bias reporting, and quantities that should remain backend diagnostics.

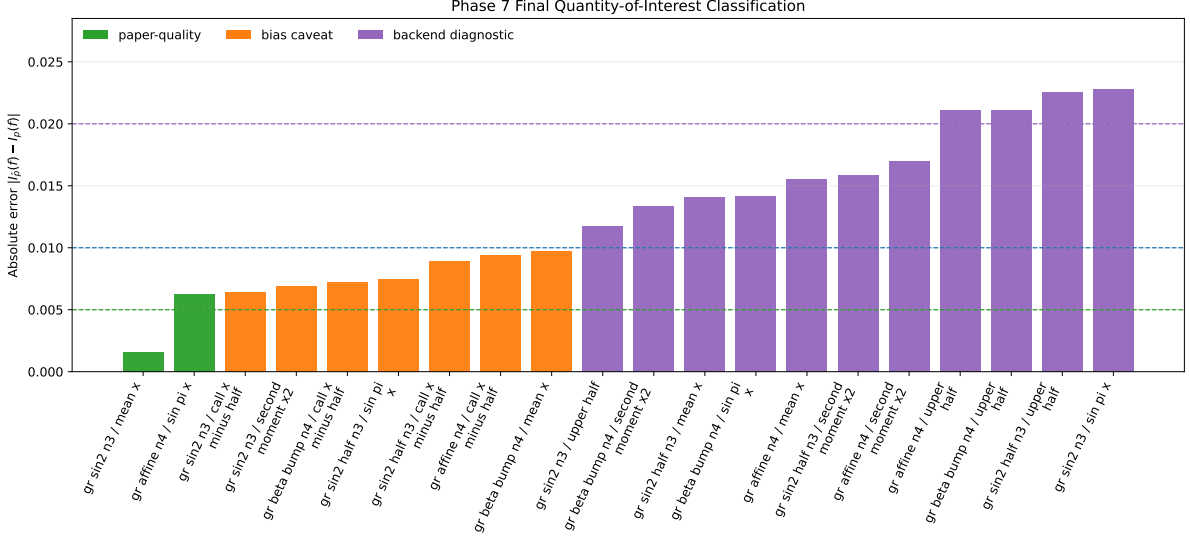


Figure 4: Final Phase 7 quantity-of-interest classification. Bar height is the absolute functional error $|I_{\hat{p}}(f) - I_p(f)|$. Color reports the final decision class: paper-quality, paper-quality with explicit bias caveat, or backend diagnostic.

The resulting publication narrative is therefore precise. The hardware evidence supports one global distribution claim, namely the mitigated **gr_affine_n4** loader, and a small set of quantity-specific functionals. Several initially strong Phase 6 quantities degrade in Phase 7, which makes them valuable diagnostics of backend drift rather than primary numerical estimates. This distinction is central to the proposed methodology: the experimental object is the finite probability state and the algebraic functional $I_p(f)$, with reproducibility assessed at the level of the selected test function.

The final **gr_beta_bump_n4** repeat removes the remaining pending replication entries. Its global distribution error remains diagnostic (TVD = 0.057598), but two smooth or payoff-like functionals remain below 10^{-2} : $I_p(x)$, with absolute error 0.009738, and $I_p((x - 1/2)_+)$, with absolute error 0.007214. The upper-half event does not replicate and is rejected in this run, with absolute error 0.021095.

The run-to-run comparison in Figure 5 explains why the classification is conservative. Only two selected functionals improve from Phase 6 to Phase 7, while most errors increase when the measurements are repeated at higher shot count. This pattern points to backend drift and functional-dependent bias rather than to simple sampling noise.

5.11 Cross-backend validation on ibm_kingston

The first Phase 8 experiment repeats the strongest Phase 7 result on a second backend. We selected the mitigated **gr_affine_n4** direct-distribution campaign because it was the only global paper-quality distribution claim on **ibm_fez**. The **ibm_kingston** backend has the same $\{cz, id, rz, sx, x\}$ basis family, but a different calibration state, so it provides a clean cross-backend control without changing the circuit family.

The Kingston run strengthens the main direct-distribution claim. Relative to Fez, it improves TVD, Hellinger distance, classical fidelity, and Euclidean distance, while keeping the maximum cell deviation below 10^{-2} . Table 24 shows that Kingston also reduces the absolute error for every tested functional. The upper-half event remains diagnostic, but its error falls



Figure 5: Change in absolute quantity-of-interest error from Phase 6 to Phase 7. Negative bars indicate improved or unchanged error; positive bars indicate degradation in the higher-shot repeat.

Backend	TVD	Hellinger	F_{cl}	Max. cell dev.
ibm_fez	0.028017	0.023874	0.998860	0.008197
ibm_kingston	0.021823	0.019060	0.999274	0.009195

Table 23: Cross-backend Phase 8 validation of mitigated **gr_affine_n4**. Both runs use 8192 shots, optimization level 2, and the same dynamical-decoupling and twirling options.

from 0.021078 to 0.014486. The strongest paper-quality functional, $I_p(\sin(\pi x))$, improves from 0.006274 to 0.004731 and retains bootstrap interval coverage.

Test function f	Fez error	Kingston error	Change
$(x - 1/2)_+$	0.009374	0.004483	-0.004891
x	0.015535	0.007235	-0.008300
x^2	0.017012	0.008438	-0.008574
$\sin(\pi x)$	0.006274	0.004731	-0.001543
$1_{x \geq 1/2}$	0.021078	0.014486	-0.006592

Table 24: Quantity-of-interest error changes for mitigated **gr_affine_n4** when moving from **ibm_fez** to **ibm_kingston**. Negative change means Kingston improves the absolute error.

This result changes the status of the affine loader from a single-backend success to cross-backend evidence. It also gives a useful warning for the next Phase 8 submissions: the second backend can materially improve the same compiled-depth and same-two-qubit-count circuit, so the remaining diagnostic families should be tested on Kingston only when the scientific question is backend dependence, not random repetition.

5.12 Phase 9 integration-scaling audit

The first Phase 9 run is a backend-aware, credit-free audit on **ibm_kingston**. It does not submit QPU jobs. It transpiles the Grover–Rudolph loaders, simulates ideal shot noise, and uses the backend noise model to estimate both global distribution error and the maximum quantity-of-interest error over the selected functionals. The results are summarized in Table 25.

Case	Depth	2q gates	Noisy TVD	Max. QoI err.	Status
gr_sin2_n3	31	7	0.028317	0.012576	candidate
gr_sin2_n4	62	16	0.029796	0.013991	candidate
gr_beta_bump_n4	68	16	0.041681	0.007289	candidate
gr_affine_n4	74	16	0.028333	0.008659	candidate
gr_beta_bump_n5	162	44	0.048994	0.014625	candidate
gr_affine_n5	171	44	0.044607	0.013084	candidate
gr_sin2_n5	141	44	0.052208	0.013443	reject TVD

Table 25: First Phase 9 integration-scaling audit on **ibm_kingston**. The candidate threshold is noisy TVD below 0.05 and maximum quantity-of-interest error below 0.02, in addition to depth and two-qubit-gate limits.

The audit gives a useful first ordering. The smooth symmetric $n = 5$ law is rejected only by the noisy-TVD threshold, while its maximum functional error remains below 0.02. The affine and localized $n = 5$ laws both pass the predeclared thresholds. Among them, **gr_affine_n5** is the most natural first hardware candidate because it refines the affine law already validated across Fez and Kingston.

A focused layout and seed audit was then run for **gr_affine_n5**. The best candidate uses optimization level 2 and transpiler seed 45678. It keeps the same structural cost as the initial dry run, depth 171 and 44 two-qubit gates, but improves the backend-noise prediction to TVD 0.025125 and maximum quantity-of-interest error 0.005126. This is the configuration prepared for the first Phase 9 hardware submission; no QPU job has been submitted at this stage.

The corresponding hardware run was then executed on **ibm_kingston** with 8192 shots. The observed distribution metrics are shown in Table 26. The measured TVD is larger than the best noisy-model prediction, but it remains below the predeclared 0.05 threshold, and the maximum cell deviation remains below 10^{-2} .

Case	Grid	Depth	2q gates	TVD	Hellinger	Max. cell dev.
gr_affine_n5	32	171	44	0.033694	0.031298	0.008438

Table 26: Phase 9 **gr_affine_n5** direct-distribution hardware result on **ibm_kingston**. The run uses Runtime mitigation, 8192 shots, optimization level 2, and transpiler seed 45678.

The quantity-of-interest behavior is more selective, as summarized in Table 27. The smooth test function $\sin(\pi x)$ remains the strongest result, with absolute error 0.003601 and bootstrap interval coverage of the exact finite-law value. The call payoff is below 10^{-2} but lacks interval coverage. The mean, second moment, and upper-half event show residual bias at the 1.5×10^{-2} scale.

Test function f	Estimate	Exact	Error	CI contains exact	Status
$\sin(\pi x)$	0.640477	0.636876	0.003601	yes	strongest
$(x - 1/2)_+$	0.176748	0.185068	0.008320	no	bias caveat
x	0.605309	0.620136	0.014827	no	diagnostic
x^2	0.437745	0.453388	0.015642	no	diagnostic
$1_{x \geq 1/2}$	0.663940	0.680380	0.016439	no	diagnostic

Table 27: Phase 9 **gr_affine_n5** quantity-of-interest estimates from the measured 32-cell probability law.

This is the first positive 32-cell scaling result in the project. It supports the direct Grover–Rudolph distribution workflow beyond the 16-cell affine anchor, but it also confirms that integration claims must remain quantity-specific. The hardware result should therefore be reported as a paper-quality distribution-scaling result, with $\sin(\pi x)$ as the cleanest associated functional and the tail/moment quantities retained as backend-bias diagnostics.

The next credit-free Phase 9 audit targeted the localized 32-cell law `gr_beta_bump_n5`. This is a stress case because localized probability mass was less stable in the earlier $n = 4$ campaigns. A layout and seed search on `ibm_kingston` selected optimization level 2 and transpiler seed 67890. The prepared circuit has depth 162, 44 two-qubit gates, and two-qubit depth 42. The backend-noise prediction is TVD 0.032691 and maximum quantity-of-interest error 0.004585. This makes `gr_beta_bump_n5` the next prepared hardware candidate.

The hardware result did not confirm the optimistic noisy-model prediction. The measured TVD is 0.073298, with Hellinger distance 0.110712 and classical fidelity 0.975636. The maximum cell deviation remains moderate, 0.014583, but the global shape error is above the predeclared Phase 9 threshold. The quantity-of-interest errors also fail to show a clean functional result: $(x - 1/2)_+$ has error 0.012934, x has error 0.020770, x^2 has error 0.024336, $\sin(\pi x)$ has error 0.013662, and the upper-half event has error 0.037838; none of the bootstrap intervals contains the exact finite-law value.

Case	Depth	2q gates	TVD	Max. dev.	Decision
<code>gr_affine_n5</code>	171	44	0.033694	0.008438	paper-quality
<code>gr_beta_bump_n5</code>	162	44	0.073298	0.014583	diagnostic

Table 28: Phase 9 closing comparison for 32-cell direct Grover–Rudolph distribution loading on `ibm_kingston`.

Phase 9 therefore gives a sharp boundary. The affine law scales successfully from 16 to 32 cells under the direct distribution workflow, while the localized beta-like law does not. This is not a failure of the experimental program; it identifies localized probability structure as the next limiting regime for current hardware and mitigation settings.

5.13 Phase 10 comparison with MC/QMC sampling

Phase 10 uses the two Phase 9 32-cell hardware distributions and compares them with MC and randomized QMC sampling from the same finite law, using the same sample count $M = 8192$. The distribution-level comparison is shown in Table 29. The QPU TVD for `gr_affine_n5` is above the MC median but still at the same order of magnitude. The `gr_beta_bump_n5` TVD is far outside the MC sampling scale. The QMC baseline is much sharper because it uses the exact finite inverse CDF; it is therefore a reference baseline once p is known. Figure 6 visualizes the same comparison: blue bars are the QPU empirical TVD, orange bars are the MC median TVD, and green bars are the randomized QMC median TVD.

Case	QPU	MC med.	QMC med.	Rank MC	Rank QMC
<code>gr_affine_n5</code>	0.033694	0.023578	0.000634	0.997	1.000
<code>gr_beta_bump_n5</code>	0.073298	0.020190	0.000632	1.000	1.000

Table 29: Phase 10 distribution-level sampling comparison. Percentile ranks report the fraction of MC/QMC trials whose TVD is no larger than the QPU TVD.

The quantity-of-interest comparison is more nuanced. For `gr_affine_n5`, $I_p(\sin(\pi x))$ remains the only estimate at the MC sampling scale: its QPU error is 0.003601, compared with MC RMSE 0.003294. All other tested quantities have QPU errors between 3 and 5 times the

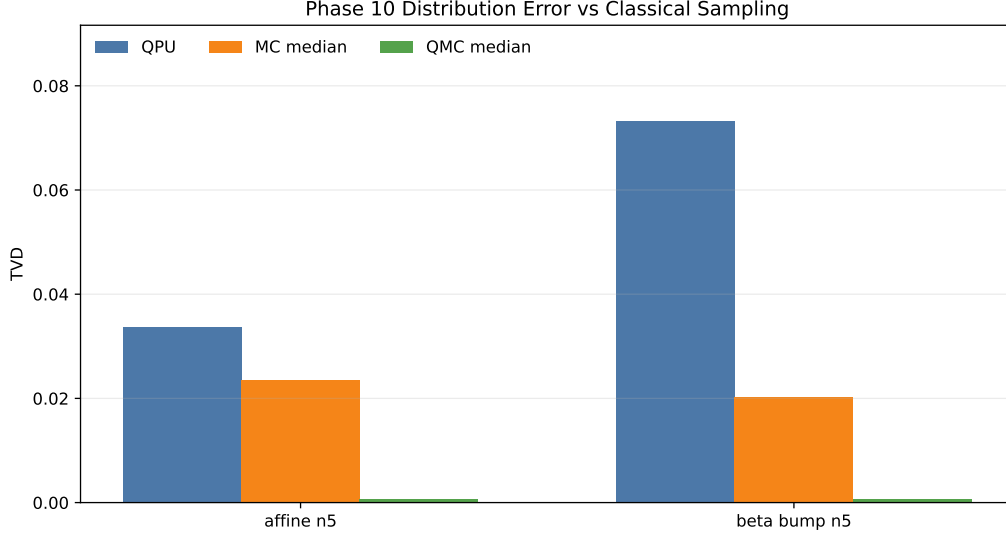


Figure 6: Phase 10 distribution-level comparison against MC and QMC sampling from the same finite probability laws.

corresponding MC RMSE. For `gr_beta_bump_n5`, every tested quantity is substantially worse than MC, confirming that this case should be reported as a localized-profile stress diagnostic. Table 30 lists representative rows. Figure 7 uses color to encode the decision status: green denotes MC-competitive, orange denotes hardware-bias diagnostic, red denotes MC-dominated, and gray denotes the MC RMSE baseline.

Case	Test function f	QPU error	MC RMSE	QMC RMSE	QPU/MC
<code>gr_affine_n5</code>	$\sin(\pi x)$	0.003601	0.003294	≈ 0	1.09
<code>gr_affine_n5</code>	$(x - 1/2)_+$	0.008320	0.001882	0.000019	4.42
<code>gr_affine_n5</code>	x	0.014827	0.002870	0.000037	5.17
<code>gr_affine_n5</code>	$1_{x \geq 1/2}$	0.016439	0.005177	0.000057	3.18
<code>gr_beta_bump_n5</code>	$(x - 1/2)_+$	0.012934	0.000413	0.000017	31.34
<code>gr_beta_bump_n5</code>	$\sin(\pi x)$	0.013662	0.002793	0.000013	4.89
<code>gr_beta_bump_n5</code>	$1_{x \geq 1/2}$	0.037838	0.003464	0.000021	10.92

Table 30: Phase 10 quantity-of-interest comparison against MC/QMC sampling from the same finite probability law with $M = 8192$. The near-zero QMC RMSE for $\sin(\pi x)$ in the affine case is a finite-grid effect of inverse-CDF QMC, and should not be interpreted as a full cost model.

5.14 Phase 11 dual angle-structure decisions

Phase 11 combines the structural angle classes with the Phase 9 and Phase 10 evidence. The resulting table is not a new hardware result; it is a credit-free policy for what should be attempted next. The most important rows are listed in Table 31. They show that the only current paper-quality direct integral remains the 32-cell affine law with $f(x) = \sin(\pi x)$. Low-angle quantities for the $\sin^2(\pi x)$ profiles are the natural simulator candidates for amplified QAE. Localized beta-like tail quantities are deferred because Phase 10 already shows MC-dominated hardware error.

The practical implication is that angle degree is a filter, not a verdict. Low angle degree is sufficient to justify simulator-level amplified-QAE studies, but it does not override measured

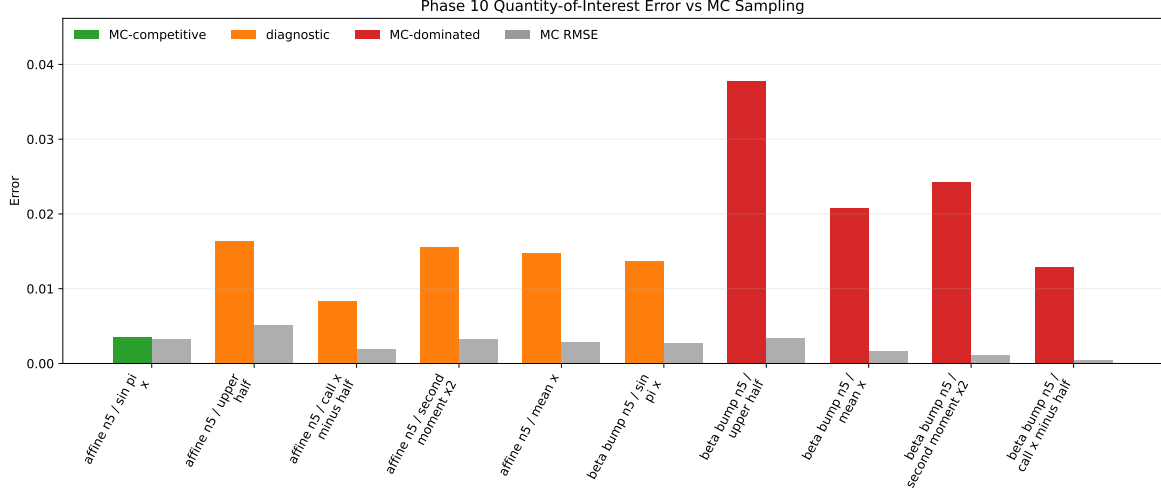


Figure 7: Phase 10 quantity-of-interest comparison. Colored bars are QPU absolute errors, and gray bars are MC RMSE at the same sample count.

Case	Test function f	Profile class	f class	Recommendation
gr_affine_n5	$\sin(\pi x)$	$\mathcal{G}_5^{(5)}$	$\mathcal{G}_5^{(4)}$	paper-quality direct
gr_sin2_n3	$1_{x \geq 1/2}$	$\mathcal{G}_3^{(2)}$	$\mathcal{G}_3^{(1)}$	simulate QAE
gr_sin2_n3	$\sin(\pi x)$	$\mathcal{G}_3^{(2)}$	$\mathcal{G}_3^{(2)}$	simulate QAE
gr_sin2_n4	$1_{x \geq 1/2}$	$\mathcal{G}_4^{(2)}$	$\mathcal{G}_4^{(1)}$	simulate QAE
gr_affine_n5	$1_{x \geq 1/2}$	$\mathcal{G}_5^{(5)}$	$\mathcal{G}_5^{(1)}$	simulator only
gr_beta_bump_n5	$1_{x \geq 1/2}$	$\mathcal{G}_5^{(5)}$	$\mathcal{G}_5^{(1)}$	defer QPU repeat
gr_beta_bump_n5	$(x - 1/2)_+$	$\mathcal{G}_5^{(5)}$	$\mathcal{G}_5^{(5)}$	defer QPU repeat

Table 31: Phase 11 dual angle-structure decisions. The profile class concerns the law-defining profile F interpreted as an amplitude function; the function class concerns the test function f .

hardware bias. High angle degree does not forbid direct Grover–Rudolph integration, but it is a warning against immediately recasting the same quantity as an amplified-QAE amplitude.

5.15 Phase 12 local QAE-amplitude audit

Phase 12 constructs the actual amplitude-estimation circuits for the Phase 11 QAE candidates. The augmented state preparation loads the finite law p on the grid register and the test function f on one objective qubit, so that the marked probability is $a = I_p(f)$. The exact statevector probabilities match $p_k(a)$ to numerical precision; the maximum discrepancy across the audit is 2.4×10^{-14} . The relevant outcome is therefore not logical correctness, but schedule informativeness and transpiled cost.

Case	Test function f	$a = I_p(f)$	Max. k	2q gates	Recommendation
gr_sin2_n3	$1_{x \geq 1/2}$	0.500000	2	35	degenerate
gr_sin2_n4	$1_{x \geq 1/2}$	0.500000	2	87	degenerate
gr_affine_n5	$1_{x \geq 1/2}$	0.680380	1	113	$k = 1$ depth watch
gr_sin2_n3	$\sin(\pi x)$	0.848596	2	86	simulator candidate

Table 32: Phase 12 local audit of augmented QAE amplitude circuits. The two upper-half $\sin^2(\pi x)$ cases have low angle degree but are not informative for QAE because $a = 1/2$, hence $p_k(a) = 1/2$ for all k .

This result sharpens the Phase 11 policy. Angle class is necessary for candidate selection, but it must be combined with the scalar amplitude value and the amplified schedule. The only non-degenerate local QAE candidate in this audit is **gr_sin2_n3** with $f(x) = \sin(\pi x)$, which reaches $k = 2$ with 86 two-qubit gates after local **cz**-basis transpilation. The 32-cell affine upper-half pair has useful amplification contrast, but only passes through $k = 1$ under the current thresholds and is therefore not a QPU candidate yet.

5.16 Phase 13 backend-aware noisy QAE audit

The only non-degenerate Phase 12 QAE candidate, **gr_sin2_n3** with $f(x) = \sin(\pi x)$, was then tested with backend-derived noisy simulation. The audit was run on both **ibm_kingston** and **ibm_fez**. No QPU jobs were submitted. The local basis-only circuit reaches $k = 2$ with depth 288 and 86 two-qubit gates, but the backend-aware transpilation/noise audit rejects a full $K = \{0, 1, 2\}$ hardware schedule.

Backend	k	Depth	2q gates	Expected p_k	Noisy p_k	Dev.
ibm_kingston	0	68	20	0.848596	0.494873	0.353723
ibm_kingston	1	222	63	0.131990	0.491211	0.359221
ibm_fez	0	60	18	0.848596	0.501953	0.346643
ibm_fez	1	292	83	0.131990	0.477783	0.345793

Table 33: Phase 13 backend-aware noisy audit for the augmented QAE candidate **gr_sin2_n3**, $f(x) = \sin(\pi x)$. Both backends collapse the marked probabilities toward roughly one half, and $k = 2$ is not retained under the current cost thresholds.

This is decisive negative evidence before spending credits. The candidate is logically correct and structurally meaningful, but current backend noise makes the amplified probability model unusable. The project should therefore keep augmented QAE in the simulator/noisy-audit track and continue using direct Grover–Rudolph distribution sampling as the hardware-facing workflow.

5.17 Phase 14 local scaling audit for direct integration

The next local audit compares direct Grover–Rudolph sampling from $n = 4$ to $n = 6$ before any backend decision. Table 34 summarizes the best local **cz**-basis transpilation over the tested optimization levels and seeds. The class column refers to the positive profile after rescaling to $[0, 1]$, used only as an angle-structure diagnostic.

The local result is encouraging but deliberately limited. Moving from 32 to 64 midpoint cells roughly doubles the two-qubit count from 30 to 62 in this local basis-only transpilation, while the worst midpoint discretization error across the tested quantities remains below 3.3×10^{-3} for $n = 4$ and below 2.1×10^{-4} for the $n = 6$ rows. Thus no $n = 6$ direct-sampling candidate is rejected locally. However, Phase 9 already showed that localized profiles can fail on hardware despite moderate circuit size. The proper next step is therefore a backend-aware layout/noise audit for these $n = 6$ rows, followed by at most one carefully selected QPU campaign.

That backend-aware audit was performed for the three $n = 6$ rows on both **ibm_kingston** and **ibm_fez**, still without submitting QPU jobs. Table 35 reports the best transpiled candidate for each backend and the noisy simulation diagnostics.

The most coherent next hardware test is therefore not the row with the lowest single noisy quantity error, but the row that extends the existing paper-quality line of evidence. The recommended microcampaign is **gr_affine_n6** on **ibm_kingston**. It extends the successful 32-cell affine result to 64 cells and remains below the noisy TVD and quantity-of-interest error thresholds. The localized beta-like row remains a stress diagnostic because Phase 9 already

Case	n	Grid	Class	Depth	2q gates
gr_sin2_n4	4	16	$\mathcal{G}_4^{(4)}$	52	14
gr_sin2_n5	5	32	$\mathcal{G}_5^{(4)}$	102	30
gr_sin2_n6	6	64	$\mathcal{G}_6^{(6)}$	200	62
gr_affine_n4	4	16	$\mathcal{G}_4^{(4)}$	69	14
gr_affine_n5	5	32	$\mathcal{G}_5^{(5)}$	141	30
gr_affine_n6	6	64	$\mathcal{G}_6^{(6)}$	287	62
gr_beta_bump_n4	4	16	$\mathcal{G}_4^{(4)}$	61	14
gr_beta_bump_n5	5	32	$\mathcal{G}_5^{(5)}$	125	30
gr_beta_bump_n6	6	64	$\mathcal{G}_6^{(6)}$	253	62

Table 34: Phase 14 local direct-sampling scaling audit. All rows pass the local cost and discretization thresholds and are therefore candidates for a later backend-aware audit. This table does not imply QPU submission.

Backend	Case	Depth	2q gates	Noisy TVD	Max QoI err.	Status
ibm_kingston	gr_sin2_n6	294	103	0.057414	0.024009	candidate
ibm_kingston	gr_affine_n6	361	103	0.046682	0.014242	candidate
ibm_kingston	gr_beta_bump_n6	338	103	0.047766	0.009092	candidate
ibm_fez	gr_sin2_n6	299	103	0.069657	0.034643	hold
ibm_fez	gr_affine_n6	361	103	0.048864	0.028890	hold
ibm_fez	gr_beta_bump_n6	337	103	0.059496	0.017425	candidate

Table 35: Phase 14 backend-aware direct-sampling audit for the $n = 6$ candidates. The status is a pre-submission decision from noisy simulation and layout-aware transpilation; no QPU jobs are submitted by this audit.

showed that this family can look reasonable in pre-submission audits while failing as a clean integration result on hardware.

The selected **gr_affine_n6** microcampaign was then executed on **ibm_kingston** with 8192 shots, dynamical decoupling, and twirling. The measured distribution has TVD 0.058478 and maximum cell deviation 0.004792, using a depth-361 circuit with 103 two-qubit gates. This passes the Phase 14 distribution rule, but the quantity-of-interest estimates in Table 36 show a strong functional dependence.

Quantity of interest	Estimate	Exact	Error	95% CI contains exact	Status
$\sin(\pi x)$	0.637435	0.636684	0.000752	yes	paper-quality
$(x - 1/2)_+$	0.168151	0.185112	0.016961	no	diagnostic
x	0.586767	0.620224	0.033457	no	diagnostic
x^2	0.419880	0.453537	0.033657	no	diagnostic
$1_{x \geq 1/2}$	0.630249	0.680380	0.050131	no	diagnostic

Table 36: Phase 14 quantity-of-interest estimates from the 64-cell affine Grover–Rudolph distribution measured on **ibm_kingston**.

Compared with the earlier 32-cell affine result, the distribution TVD increases from 0.033694 to 0.058478. Nevertheless, the most robust integration quantity, $I_p(\sin(\pi x))$, improves from absolute error 0.003601 to 0.000752 and is statistically compatible with the exact finite-law value. The tail and moment quantities become worse and remain backend-bias diagnostics. Thus the correct Phase 14 claim is not a uniform improvement from $n = 5$ to $n = 6$, but a stronger paper-quality result for the smooth quantity $I_p(\sin(\pi x))$ on a 64-cell law.

5.18 Phase 15 centered Gaussian finite-law test

The Phase 15 backend-aware audit selected the broad centered Gaussian-like profile $\sigma = 0.25$ as the best alternative $n = 6$ probability law on `ibm_kingston`. The selected circuit had depth 193 and 66 two-qubit gates, substantially below the affine $n = 6$ circuit. The noisy pre-submission model predicted TVD 0.044257 and maximum quantity-of-interest error 0.012389.

The QPU execution gives TVD 0.070511 and maximum cell deviation 0.005211. Thus the distribution shape is not globally better than the affine $n = 6$ result, but the quantity-of-interest behaviour is complementary, as shown in Table 37.

Quantity of interest	Estimate	Exact	Error	95% CI contains exact	Status
$1_{x \geq 1/2}$	0.496338	0.500000	0.003662	yes	validation
x	0.496075	0.500000	0.003925	yes	validation
x^2	0.304913	0.298362	0.006552	no	diagnostic
$(x - 1/2)_+$	0.098776	0.090369	0.008407	no	diagnostic
$\sin(\pi x)$	0.738914	0.782703	0.043789	no	diagnostic

Table 37: Phase 15 quantity-of-interest estimates from the 64-cell centered Gaussian-like Grover–Rudolph distribution measured on `ibm_kingston`.

This closes the direct finite-law block with a deliberately non-uniform conclusion. The affine $n = 6$ law remains the best result for the smooth quantity $I_p(\sin(\pi x))$, with error 0.000752. The centered Gaussian-like law gives better estimates for the upper-half mass and the mean, both with interval coverage, but it fails badly for $I_p(\sin(\pi x))$. The relevant lesson is therefore not that one probability law is uniformly superior; it is that the hardware bias couples strongly to both the prepared law p and the chosen test function f .

5.19 Bridge to quadrature-rule QAE integration

The preceding phases study finite-law integration after probability-law preparation. This workflow estimates $I_p(f)$ from measured frequencies and is well suited to testing Grover–Rudolph state preparation on hardware. The angle-structure paper, however, formulates numerical integration differently: the integrand g is encoded as an amplitude, and the numerical rule is one of the left Riemann, midpoint, right Riemann, or Simpson quadrature schemes. The next experimental block should therefore return to that formulation and measure

$$I[g] - \widehat{Q}_n^{(p)}[g] = (I[g] - Q_n^{(p)}[g]) + (Q_n^{(p)}[g] - \widehat{Q}_n^{(p)}[g]),$$

separating discretization error from QAE/MLAE estimation error.

5.20 Phase 15b quadrature-rule QAE hardware results

The `ibm_kingston` midpoint-quadrature microcampaign completed with all ten submitted jobs in the DONE state. Table 38 summarizes the grouped MLAE estimates. The reported quadrature value $Q_n[g]$ is the exact midpoint rule value on the selected grid; the true value $I[g]$ is the high-accuracy classical reference integral.

The two calibration functions are strong positive evidence for the restored Workflow-A implementation. The g_2 result is particularly clean: $\widehat{Q}_n[g] = 0.500140$, with total error 1.4×10^{-4} , using the full $K = \{0, 1, 2\}$ schedule. The g_1 result also remains at the 10^{-3} level. These two cases reproduce, in the quadrature-rule language of the angle-structure paper, the earlier conclusion that the low-degree calibration hierarchy is hardware-stable for shallow schedules.

The two stress functions give complementary information. The shifted affine case has no discretization error on this rule, but shows a visible hardware estimation bias of 0.010640,

Case	$Q_n[g]$	$I[g]$	$\hat{Q}_n[g]$	Est. err.	Total err.
<code>g1_n2_midpoint</code>	0.500000	0.500000	0.498828	0.001172	0.001172
<code>g2_n2_midpoint</code>	0.500000	0.500000	0.500140	0.000140	0.000140
<code>affine_shifted_n2_midpoint</code>	0.395000	0.395000	0.405640	0.010640	0.010640
<code>smooth_bump_n3_midpoint</code>	0.033974	0.033333	0.034902	0.000928	0.001568

Table 38: Phase 15b midpoint-quadrature MLAE estimates from `ibm_kingston`. “Est. err.” is $|\hat{Q}_n[g] - Q_n[g]|$; “Total err.” is $|\hat{Q}_n[g] - I[g]|$.

dominated by the $k = 1$ probability shift. It should therefore remain a watch case. The smooth bump has a nonzero midpoint discretization error, 0.000640, but its MLAE estimation error is only 0.000928, giving total error 0.001568. This is important because it shows that the quadrature-QAE workflow is not limited to amplitudes near $1/2$; it can also estimate a low-amplitude smooth integral when the circuit remains within the audited depth and two-qubit-gate regime.

The circuit-level probabilities also support this interpretation. For `g1` and `g2`, all amplified probabilities remain close to the ideal value $1/2$. For `affine_shifted`, the $k = 1$ probability is 0.765625 instead of 0.796478, explaining the upward bias in the inverted amplitude. For `smooth_bump`, the individual amplified probabilities show larger deviations, but the likelihood combination still recovers the underlying small quadrature value accurately enough for this first microcampaign.

The same ten-circuit campaign was then repeated on `ibm_aachen`, using the IBM Quantum Credits Europe instance. This was a deliberately clean cross-backend test: the mathematical cases, quadrature rule, schedules, shots, optimization level, dynamical decoupling, and twirling settings were kept fixed. At submission time `ibm_aachen` had substantially better average two-qubit and readout errors than `ibm_kingston`. The grouped MLAE estimates are shown in Table 39. The hardware outcome, however, was not uniformly better than the Kingston run, as shown in Table 40.

Case	$Q_n[g]$	$I[g]$	$\hat{Q}_n[g]$	Est. err.	Total err.
<code>g1_n2_midpoint</code>	0.500000	0.500000	0.494726	0.005274	0.005274
<code>g2_n2_midpoint</code>	0.500000	0.500000	0.503518	0.003518	0.003518
<code>affine_shifted_n2_midpoint</code>	0.395000	0.395000	0.502980	0.107980	0.107980
<code>smooth_bump_n3_midpoint</code>	0.033974	0.033333	0.032060	0.001914	0.001274

Table 39: Phase 15b midpoint-quadrature MLAE estimates from `ibm_aachen`. The campaign is the same as in Table 38; only the backend and IBM Quantum instance are changed.

Case	Kingston total err.	Aachen total err.	Better backend
<code>g1_n2_midpoint</code>	0.001172	0.005274	Kingston
<code>g2_n2_midpoint</code>	0.000140	0.003518	Kingston
<code>affine_shifted_n2_midpoint</code>	0.010640	0.107980	Kingston
<code>smooth_bump_n3_midpoint</code>	0.001568	0.001274	Aachen

Table 40: Cross-backend Phase 15b comparison between the original `ibm_kingston` run and the repeated `ibm_aachen` run. Both runs used the same midpoint-quadrature MLAE micro-campaign and 2048 shots per circuit.

This repeat is important negative and diagnostic evidence. The calibration cases remain ac-

ceptable on `ibm_aachen`, but both are worse than on `ibm_kingston`. The smooth low-amplitude bump improves slightly on `ibm_aachen`, from total error 0.001568 to 0.001274. In contrast, the shifted affine case fails as a reliable MLAE estimate on `ibm_aachen`: the $k = 1$ hardware probability is 0.454102 instead of the expected 0.796478, and the inferred amplitude moves to 0.502980 rather than the target 0.395000. Thus a backend with better average calibration metrics can still be worse for a specific amplified circuit. Backend selection must therefore remain circuit-specific and evidence-based, not based only on global average error summaries.

5.21 Phase 15c comparison with Monte Carlo and quasi-Monte Carlo

The Phase 15b quadrature-QAE estimates were then compared with classical Monte Carlo and randomized quasi-Monte Carlo estimators for the same continuous integrals $I[g] = \int_0^1 g(x) dx$. This comparison is intentionally conservative for the quantum workflow: MC and QMC sample the function g directly, while the QPU estimates inherit the full cost of state preparation, amplification, transpilation, readout, and backend noise. The purpose is not to claim an immediate advantage, but to identify where the angle-structured workflow already beats the $M^{-1/2}$ MC scale and where hardware errors still dominate.

Table 41 reports the comparison with the same sample budget as the total number of hardware shots used by each MLAE group. The repository also stores the matched oracle-query-budget comparison, where the classical sample count is set to the number of oracle applications $\sum_{k \in K} (2k + 1)M_k$. The QMC values should be read as a strong deterministic-sequence baseline rather than as a claim about hardware cost: for the small grids and smooth functions used here, randomized van der Corput sampling can integrate some trigonometric cases almost exactly.

Backend	Case	QPU err.	MC RMSE	QMC RMSE	QPU/MC	QPU/QMC
kingston	g1_n2_midpoint	0.001172	0.005633	0.000070	0.21	16.84
kingston	g2_n2_midpoint	0.000140	0.004462	$< 10^{-6}$	0.03	1032.77
kingston	affine_shifted_n2_midpoint	0.010640	0.002638	0.000040	4.03	269.08
kingston	smooth_bump_n3_midpoint	0.001568	0.000387	$< 10^{-6}$	4.05	104196.06
aachen	g1_n2_midpoint	0.005274	0.005378	0.000069	0.98	76.92
aachen	g2_n2_midpoint	0.003518	0.004368	$< 10^{-6}$	0.81	26095.44
aachen	affine_shifted_n2_midpoint	0.107980	0.002664	0.000040	40.53	2712.19
aachen	smooth_bump_n3_midpoint	0.001274	0.000381	$< 10^{-6}$	3.34	85387.21

Table 41: Phase 15c comparison of Phase 15b midpoint-quadrature MLAE hardware estimates with MC/QMC estimators of the continuous integral. The sample budget is matched to the total number of hardware shots in each grouped MLAE estimate.

This table gives a sharper interpretation of the Phase 15b result. On `ibm_kingston`, the calibration functions g_1 and g_2 are not only accurate in absolute terms; they also beat the MC RMSE at the same shot budget. This is the cleanest evidence that the angle-structured workflow can produce a useful amplification-based integration estimate on current hardware for the lowest-degree classes. In contrast, the shifted affine and smooth bump cases do not beat MC at the same shot budget. Their errors are dominated by backend and amplification effects rather than by the classical sampling scale. The `ibm_aachen` repeat confirms the same qualitative picture, but with worse constants for g_1 , g_2 , and especially the shifted affine stress case.

The QMC comparison is deliberately demanding. For these one-dimensional smooth tests, randomized low-discrepancy sampling is extremely effective, so it should be treated as a high bar for near-term quantum integration. This motivates the next experimental direction: scaling the angle-structured quadrature workflow only where the audited circuit depth and two-qubit count remain controlled, while using MC/QMC as mandatory baselines for every proposed hardware microcampaign.

5.22 Phase 16 backend-aware scaling decision for Workflow A

Phase 16 turns the previous conclusion into an operational rule and aligns it with the angle-structure characterization of Chinesta–Falcó–Falcó–Pomares. The relevant invariant is not only the smoothness of g , but the multilinear degree of the QAE angle table

$$\Theta_g(b) = 2 \arcsin \sqrt{g(x_{i(b)})}.$$

For each audited row we therefore compute the class $\mathcal{G}_n^{(d)}$, using Möbius inversion/Walsh–Hadamard coefficients with tolerance 10^{-10} . Starting from the complete Phase 15 quadrature-QAE audit, a candidate is promoted only if:

- (i) its sampled integrand has a recorded angle class $\mathcal{G}_n^{(d)}$;
- (ii) the maximum transpiled depth is at most 450;
- (iii) the maximum two-qubit-gate count is at most 150;
- (iv) the noisy-simulation total error is below 0.01;
- (v) the noisy-simulation total error is no larger than the matched MC RMSE at the same shot budget.

Randomized QMC is computed and reported for every cost-controlled row, but is used as a strict reference rather than as a hard veto. In this one-dimensional smooth setting it is often nearly exact and therefore represents a demanding classical benchmark.

Case	Class and K	Depth	2q gates	Pred. QPU err.	MC RMSE	QPU/MC	Decision
g2	$\mathcal{G}_5^{(2)}, K = \{0, 1\}$	339	110	0.000293	0.005524	0.05	promote
g2	$\mathcal{G}_3^{(2)}, K = \{0, 1, 2\}$	172	50	0.000391	0.004511	0.09	promote
g1	$\mathcal{G}_2^{(1)}, K = \{0, 1\}$	53	15	0.000488	0.005524	0.09	promote
oscillatory_smooth	$\mathcal{G}_2^{(1)}, K = \{0, 1, 2\}$	73	18	0.000251	0.002429	0.10	promote
g2	$\mathcal{G}_4^{(2)}, K = \{0, 1\}$	184	56	0.001904	0.005524	0.34	promote
g1	$\mathcal{G}_5^{(1)}, K = \{0, 1\}$	317	111	0.003760	0.005524	0.68	promote
oscillatory_smooth	$\mathcal{G}_4^{(3)}, K = \{0, 1\}$	253	75	0.002930	0.002975	0.98	marginal

Table 42: Representative Phase 16 shot-budget decision rows. The predicted QPU error is the noisy-simulation total error from the Phase 15 audit when available. MC RMSE uses the same total number of shots as the grouped MLAE estimate.

The first hardware action selected from this table is deliberately minimal: **g2** with $n = 5$, the midpoint rule, and $K = \{0, 1\}$. This extends the validated g_2 quadrature experiment from the $n = 2$ hardware calibration to a 32-point grid while keeping the predicted QPU error well below the matched MC RMSE. It is also structurally motivated: g_2 remains in $\mathcal{G}_n^{(2)}$ as n grows, whereas the shifted affine and smooth bump stress cases move to higher or full angle degree on refined grids. Thus regularity alone is not the selection criterion; the decisive quantity is the degree of $\Theta_g|X_n$, as predicted by the encoding-complexity theory. The deeper $K = \{0, 1, 2\}$ version of the same case has an excellent noisy prediction, but is rejected by the depth threshold. This is the intended behavior of the Phase 16 rule: do not spend credits on a circuit whose mathematical promise is outweighed by transpiled cost.

5.23 Phase 17 Grover–Rudolph angle precision versus shot noise

The AIMS Grover–Rudolph stability theorem gives the precise motivation for a complementary local test for Workflow B. Theorem 4.1 of [3] states that if each Grover–Rudolph angle is perturbed by at most η , then the prepared probability law changes by at most $\min(1, n\eta)$ in TVD.

Corollary 5.1 of the same paper combines this deterministic angular term with multinomial shot noise; for S shots and confidence $1 - \delta$, the form relevant here is

$$\text{TVD}(p, \hat{p}) \leq \min \left(1, \frac{n\pi}{2^{b+1}} + \sqrt{\frac{2^n \log(2/\delta)}{2S}} \right),$$

when the Grover–Rudolph angles are quantized to a b -bit grid in the physical R_y convention. Phase 17 implements exactly this separation without QPU access. For each finite law, the exact physical R_y angles are quantized to a b -bit grid; the resulting distribution is reconstructed from the dyadic tree and compared with the target law in TVD. We then compare this deterministic TVD with the empirical TVD produced by multinomial sampling with exact angles, and with the combined TVD obtained by sampling from the quantized-angle law.

Case	n	Angle TVD	Shot TVD	Combined TVD	Angle/shot
gr_sin2_n3	3	0.003632	0.014680	0.014857	0.25
gr_sin2_n4	4	0.006534	0.021382	0.022440	0.31
gr_affine_n5	5	0.006485	0.034008	0.034768	0.19
gr_beta_bump_n5	5	0.005287	0.028749	0.029384	0.18
gr_gaussian_sigma025_n6	6	0.005432	0.047425	0.047892	0.11

Table 43: Phase 17 local Grover–Rudolph angle-precision diagnostic at $b = 8$ physical-angle bits and 4096 shots, with 1000 multinomial trials.

The result in Table 43 matches the qualitative prediction of Theorem 4.1 and Corollary 5.1 of [3]. Even at $b = 8$, the deterministic angle-quantization TVD is below the finite-shot TVD for all tested profiles. At $b = 16$, the angle TVD drops to approximately 2×10^{-5} , and at $b = 32$ it is around 4×10^{-10} . Therefore, for the present Grover–Rudolph distribution experiments, moderate classical angle precision is not the limiting factor. The limiting practical terms are finite shots, backend noise, layout-dependent compilation, and mitigation behaviour.

5.24 Phase 17b backend check against the AIMS diagnostic

The local Phase 17 diagnostic can be used as a backend check by comparing the measured QPU TVD with the local finite-shot prediction for the same number of shots. This is not a new mathematical bound on the hardware run; rather, it is a decision diagnostic. If the backend TVD lies below the local Phase 17 95-percentile, the run is classified as shot-compatible. If it is above that scale but not more than twice the local value, the backend contribution is visible. Larger deviations are classified as backend-dominated and should be reported as hardware/noise diagnostics rather than as clean finite-law integration evidence.

Case	Backend	Shots	QPU TVD	Local shot mean	Local shot p95	Status
gr_affine_n5	ibm_kingston	8192	0.033694	0.023945	0.029739	backend-visible
gr_gaussian_sigma025_n6	ibm_kingston	8192	0.070511	0.033800	0.039105	backend-visible
gr_sin2_n4	ibm_kingston	8192	0.038898	0.015321	0.021492	backend-visible
gr_beta_bump_n5	ibm_kingston	8192	0.073298	0.020344	0.025889	backend-dominated
gr_sin2_n3	ibm_fez	8192	0.056027	0.010274	0.015731	backend-dominated

Table 44: Phase 17b comparison between existing QPU distribution measurements and the local Phase 17 finite-shot TVD prediction at $b = 16$. The local angle-quantization TVD is already negligible at this precision, so excess TVD is attributed to backend, layout, compilation, and mitigation effects.

Table 44 shows that none of the existing backend runs is purely shot-compatible. The closest case is **gr_affine_n5**, whose QPU TVD is only moderately above the local shot-noise

envelope. The new smooth grid-refined control `gr_sin2_n4` is also backend-visible: its depth-62, 16-two-qubit-gate direct distribution circuit remains controlled, but the measured TVD 0.038898 is about 2.54 times the local finite-shot mean and exceeds the local 95-percentile 0.021492. The localized beta-bump profile and the previously measured `gr_sin2_n3` run are backend-dominated under this criterion. This confirms that Phase 17b is useful as a rejection and interpretation rule before considering any amplified QAE circuit.

5.25 Phase 18 non-degenerate oscillatory angle-class test

The next test returns to Workflow A and uses the angle-class hierarchy to select a stress case. The original `oscillatory_smooth` midpoint instance belongs to $\mathcal{G}_4^{(3)}$, so the natural MLAE schedule from the angle-structure perspective is $K = \{0, 1, 2\}$. However, its midpoint quadrature value is exactly $a = 1/2$. This makes $p_k(a) = \sin^2((2k+1) \arcsin \sqrt{a}) = 1/2$ for all k , so the schedule is amplitude-degenerate. A dry run also rejected the full schedule on hardware: the $k = 2$ circuit reached depth 622 with 255 two-qubit gates.

We therefore introduced a minimal shifted variant

$$g(x) = 0.55 + 0.25 \sin(2\pi x) + 0.10 \cos(4\pi x).$$

On the 16-point midpoint grid this keeps the same computed angle class $\mathcal{G}_4^{(3)}$, but changes the quadrature value to $a = 0.55$. The expected amplified probabilities are no longer degenerate:

$$p_0 = 0.55, \quad p_1 = 0.352, \quad p_2 = 0.74008.$$

Because $k = 2$ remained outside the controlled hardware regime, the submitted microcampaign used the reduced schedule $K = \{0, 1\}$ on `ibm_kingston`, using the IBM Quantum Credits Program instance.

Case	k	Depth	2q gates	Expected p_k	Hardware $\hat{p}_k$
<code>oscillatory_shifted_smooth</code>	0	50	23	0.550000	0.524414
<code>oscillatory_shifted_smooth</code>	1	334	139	0.352000	0.444824

Table 45: Phase 18 hardware readout for the non-degenerate shifted oscillatory $\mathcal{G}_4^{(3)}$ quadrature instance on `ibm_kingston`.

The resulting MLAE estimate is $\hat{a} = 0.519024$. Since both the midpoint quadrature value and the continuous integral equal 0.55 for this shifted trigonometric profile, the total integration error is 0.030976. The result is therefore scientifically useful but not paper-quality as a positive hardware accuracy claim: the shift successfully removes the $a = 1/2$ degeneracy while preserving the $\mathcal{G}_4^{(3)}$ class, but the $k = 1$ circuit already shows strong backend sensitivity.

We also estimated the unsubmitted $k = 2$ circuit without spending QPU credits. The ideal value is $p_2 = 0.740080$, but the backend-derived noisy simulation gave $p_2 \simeq 0.649658$. A purely empirical extrapolation from the observed $k = 0, 1$ signed deviations instead predicts a much larger value, approximately $p_2 \simeq 0.952$. This disagreement is itself informative: the unsubmitted circuit lies outside a reliably predictable hardware regime.

The conclusion is that $k = 2$ would be beneficial in the noiseless MLAE model, but the predicted hardware behavior is unstable and model-dependent. This supports the decision to keep $k = 2$ as a diagnostic projection rather than as a hardware submission.

5.26 First amplified Grover–Rudolph readout

The first amplified test used `gr_affine_n4`, the marked event $x \geq 1/2$, and the minimal MLAE schedule $K = \{0, 1\}$. The exact marked probability is $a = 0.680380$. The $k = 0$ circuit remains

Scenario	MLAE estimate	Error to 0.55
$K = \{0, 1\}$ hardware only	0.519024	0.030976
$K = \{0, 1\}$ plus ideal $k = 2$	0.540971	0.009029
$K = \{0, 1\}$ plus noisy-simulated $k = 2$	0.527080	0.022920
$K = \{0, 1\}$ plus empirical-projected $k = 2$	0.580335	0.030335

Table 46: Phase 18 credit-free $k = 2$ projection for the shifted $\mathcal{G}_4^{(3)}$ oscillatory quadrature instance. The $k = 2$ circuit has depth 622 and 255 two-qubit gates, so it remains rejected for QPU submission.

close to the distribution-measurement regime, but the first Grover iterate is not yet stable on hardware, as shown in Table 47.

Case	k	Depth	2q gates	Expected p_k	Hardware $\hat{p}_k$
<code>gr_affine_n4</code>	0	76	16	0.680380	0.649414
<code>gr_affine_n4</code>	1	225	51	0.052765	0.173828

Table 47: First amplified Grover–Rudolph hardware readout on `ibm_fez`. The resulting MLAE estimate is $\hat{a} = 0.620525$, with absolute error 0.059854.

The $k = 0$ deviation, 0.0310, is consistent with the mitigated distribution TVD scale. The $k = 1$ deviation, 0.1211, is too large to justify submitting $k = 2$. The next step is therefore a credit-free diagnosis of `gr_affine_n4`, $k = 1$, over optimization levels, transpiler seeds, layouts, exact ideal simulation, and backend-noise simulation.

The diagnosis confirms that the logical circuit is exact:

$$p_1(a) = 0.052765.$$

The executed opt-2 circuit had depth 225 and 51 two-qubit gates. A sweep over optimization levels and transpiler seeds identifies opt-3, seed 12345 as the best repeat candidate, with depth 152, 35 two-qubit gates, and noisy-simulation probability 0.083740. This is still biased relative to the ideal value, but it is much closer than the observed hardware value 0.173828 and uses substantially fewer entangling gates.

Circuit	Depth	2q gates	Predicted/observed p_1	Abs. dev.
Executed opt-2 hardware	225	51	0.173828	0.121064
Best opt-3 diagnosis	152	35	0.083740	0.030976

Table 48: Diagnosis of the unstable `gr_affine_n4`, $k = 1$, amplified circuit. The opt-3 row is a noisy-simulation prediction, not yet a hardware result.

The opt-3, seed-12345 repeat was then submitted as a single-circuit microcampaign. It improved the measured probability from 0.173828 to 0.124023, but it did not reach the noisy prediction 0.083740 or the ideal value 0.052765. The result is summarized in Table 49.

This is useful negative evidence. The compilation improvement is real, but the hardware remains more biased than the backend-noise model predicts. The next analysis should use the already collected counts to test whether readout postprocessing can explain a substantial part of the remaining bias.

Counts-level postprocessing confirms the same conclusion. Against the ideal amplified output distribution, the opt-2 $k = 1$ circuit has TVD 0.1292, while the opt-3 repeat has TVD

Run	Opt.	Depth	2q gates	Hardware $\hat{p}_1$	Abs. dev.
Initial $k = 1$	2	225	51	0.173828	0.121064
Repeat $k = 1$	3	152	35	0.124023	0.071259

Table 49: Hardware repeat of the unstable `gr_affine_n4`, $k = 1$, amplified circuit. The repeat improves the result but does not stabilize the first Grover iterate enough to justify $k = 2$.

0.0774. This is a meaningful improvement, but still larger than the $k = 0$ distribution-level TVD 0.0472. The residual error is therefore not only a scalar estimator artifact; the amplified output distribution itself remains distorted.

A simple event-bit leakage model gives a more precise interpretation of the scalar bias. For the opt-3 repeat, fitting a two-parameter flip channel between paired cells `0xxx` and `1xxx` gives lower-to-upper leakage 0.0589 and upper-to-lower leakage 0.0000. Inverting this fitted channel moves the marked probability from 0.124023 to 0.069181, reducing the scalar deviation from 0.071259 to 0.016416. Thus most of the remaining scalar bias is compatible with leakage across the event bit defining $x \geq 1/2$, even though the full amplified distribution is still not accurate enough to justify $k = 2$. A multinomial bootstrap gives a wide 95% interval for the corrected marked probability, $[0.000000, 0.133069]$, so the fitted flip model should be treated as diagnostic evidence rather than as a reliable production correction.

We then separated final readout error from event-bit leakage using the real transpiled layouts. In the opt-2 $k = 1$ circuit, the event bit is measured as $c[0]$ from physical qubit $q[137]$; in the opt-3 repeat, it is measured as $c[0]$ from physical qubit $q[134]$. The current backend assignment probabilities for these qubits are small. Here $p(1|0)$, denoted p_{01} in the analysis files, is the probability of measuring 1 when the prepared state was 0; $p(0|1)$, denoted p_{10} , is the probability of measuring 0 when the prepared state was 1. The calibrated values are $p(1|0) = 0.00146$, $p(0|1) = 0.00708$ for $q[137]$, and $p(1|0) = 0.00098$, $p(0|1) = 0.00830$ for $q[134]$. Applying the corresponding two-state readout correction changes the opt-2 $k = 1$ marked probability from 0.173828 to 0.173849 and the opt-3 repeat from 0.124023 to 0.124199. Thus final readout calibration explains only about 0.5% of the fitted lower-to-upper leakage in the opt-2 circuit and 1.7% in the opt-3 circuit. The dominant mechanism is therefore not final measurement assignment error, but circuit-level leakage or coherent/noisy amplification dynamics before measurement.

Finally, we performed a credit-free block diagnosis of the opt-3 $k = 1$ candidate. The circuit was decomposed into checkpoints A , OA , $A^\dagger OA$, $S_0 A^\dagger OA$, and the full $AS_0 A^\dagger OA$ iterate. Table 50 reports the exact and backend-noise simulated event masses.

Block	Exact mass	Noisy mass	Depth	2q gates
A	0.680380	0.678223	74	16
OA	0.680380	0.680664	76	16
$A^\dagger OA$	0.869853	0.836914	4	0
$S_0 A^\dagger OA$	0.869853	0.823730	73	18
$AS_0 A^\dagger OA$	0.052765	0.112793	152	35

Table 50: Credit-free block diagnosis of the opt-3 `gr_affine_n4`, $k = 1$, amplified circuit on the `ibm_fez` backend-noise model.

The state-preparation block A remains accurate under the backend-noise model: the event mass changes only from 0.680380 to 0.678223. The large event-mass bias appears only after the full Grover iterate maps intermediate noise back through the final A block, moving the expected value 0.052765 to 0.112793 in noisy simulation. This supports the conclusion drawn

from hardware: the next mitigation target is amplification sensitivity, not distribution loading or final readout.

We therefore audited alternative amplification implementations before any new hardware execution. The variants changed the state-preparation decomposition (`ucry` or `direct`) and the zero-reflection implementation (`mcx`, `mcp`, or an explicit diagonal reflection). Table 51 summarizes the relevant outcomes.

Loader	Reflection	Opt.	Depth	2q gates	Noisy dev.
<code>ucry</code>	<code>mcx</code>	3	152	35	0.0312
<code>ucry</code>	diagonal	3	177	46	0.0434
<code>ucry</code>	<code>mcp</code>	3	291	80	0.1040
<code>direct</code>	<code>mcx</code>	3	2726	789	0.4131

Table 51: Credit-free amplification-variant audit for `gr_affine_n4`, $k = 1$, on the `ibm_fez` backend-noise model. The deviation is measured against the ideal $p_1 = 0.052765$.

The current `ucry` loader with the `mcx`-based zero reflection and optimization level 3 remains the best known compilation. Replacing only the zero reflection does not stabilize the amplified circuit: `mcp` and diagonal variants both increase depth, two-qubit count, and noisy deviation. The direct tree implementation is far outside the useful hardware regime for this four-qubit amplified circuit.

This motivates a less aggressive estimator for the sixteen-point grid: estimate the event probability directly from the measured Grover–Rudolph distribution, instead of applying Grover amplification. For the `gr_affine_n4` upper-half event, Table 52 compares the direct distribution estimator with the amplified readouts already collected.

Method	Estimate	Exact value	Abs. error
Mitigated distribution direct	0.671387	0.680380	0.008993
$k = 0$ event circuit	0.649414	0.680380	0.030966
Raw distribution direct	0.647461	0.680380	0.032919
MLAE inverse, $K = \{1\}$	0.639942	0.680380	0.040438
MLAE inverse, $K = \{0, 1\}$	0.620525	0.680380	0.059854

Table 52: Direct distribution estimator versus amplified estimates for the `gr_affine_n4` upper-half event. The mitigated distribution direct estimate has Wilson 95% interval $[0.656847, 0.685605]$, which contains the exact value.

The direct mitigated-distribution estimate is the best hardware estimate obtained so far for this sixteen-point event. It uses the same depth-76, 16-CZ state-preparation circuit as the distribution validation, avoids the unstable $k = 1$ Grover iterate, and remains statistically compatible with the exact value. This gives a concrete near-term workflow: for current noisy hardware, use mitigated Grover–Rudolph distribution measurements as the primary estimator for sixteen-point events, and reserve amplified QAE for families that first pass the block-level stability audit.

We also evaluated general grid functionals from the same measured Grover–Rudolph distributions. This tests whether the direct workflow supports integration tasks beyond a single event probability. Table 53 reports representative quantities of interest.

The direct-distribution workflow is therefore viable but functional-dependent. Smooth or symmetry-protected quantities remain robust, but the angle classes in Table 53 show that several apparently elementary test functions are not low-degree QAE amplitude functions on the sampled grid. For instance, x is $\mathcal{G}_4^{(3)}$ on the sixteen-point grid, while x^2 and $\sin(\pi x)$ reach

Case	Quantity of interest	Angle class	Estimate	Exact	Error
gr_affine_n4 mitigated	$1_{x \geq 1/2}$	$\mathcal{G}_4^{(1)}$	0.671387	0.680380	0.008993
gr_affine_n4 mitigated	$\sin(\pi x)$	$\mathcal{G}_4^{(4)}$	0.644002	0.637644	0.006359
gr_affine_n4 mitigated	x	$\mathcal{G}_4^{(3)}$	0.610168	0.619783	0.009615
gr_affine_n4 mitigated	x^2	$\mathcal{G}_4^{(4)}$	0.441633	0.452791	0.011158
gr_sin2_n3 raw	$1_{x \geq 1/2}$	$\mathcal{G}_3^{(1)}$	0.492188	0.500000	0.007812
gr_sin2_n3 raw	x	$\mathcal{G}_3^{(3)}$	0.494507	0.500000	0.005493
gr_sin2_half_n3 raw	x	$\mathcal{G}_3^{(3)}$	0.695557	0.701323	0.005766

Table 53: Quantity-of-interest estimates computed directly from measured Grover–Rudolph distributions. Midpoint-grid functionals $I_{\hat{p}}(f)$ are compared against $I_p(f)$ for the exact target distribution on the same grid. The angle class is the class of the test function f itself when f is interpreted as a QAE amplitude function on the corresponding n -qubit grid.

$\mathcal{G}_4^{(4)}$. Right-tail-sensitive quantities such as x , x^2 , and call-style payoffs still expose residual distribution bias in the sixteen-point affine case. This is useful diagnostic information: the remaining limitation is measurable at the distribution level, not only through amplified QAE failure.

We finally tested two lightweight classical bias corrections. First, we applied diagnostic shrinkage from the measured mitigated distribution toward the known target distribution. A 50% shrinkage halves the distribution TVD from 0.0341 to 0.0171 and approximately halves all functional errors. This quantifies the scale of the residual shape bias, but it is not a blind production correction because it uses the target distribution itself.

Second, we fitted an affine correction at the level of quantities of interest, $\hat{I}_{\text{corr}} = \alpha + \beta \hat{I}_{\text{meas}}$ and validated it by leaving one test function out. Table 54 summarizes the held-out errors.

Held-out quantity	Raw error	Calibrated error	Change
Call payoff	0.006471	0.006717	-0.000246
$ x - 1/2 $	0.003326	0.009093	-0.005767
$1_{x \geq 3/4}$	0.020783	0.014918	+0.005864
x	0.009615	0.004403	+0.005212
x^2	0.011158	0.003926	+0.007232
$\sin(\pi x)$	0.006359	0.017723	-0.011364
$1_{x \geq 1/2}$	0.008993	0.004942	+0.004051

Table 54: Leave-one-test-function-out affine calibration for the mitigated gr_affine_n4 distribution. Positive change means error reduction.

The mean absolute error improves only marginally, from 0.009529 to 0.008817, and several held-out quantities get worse. A single global affine correction is therefore not reliable enough. Any production postprocessing should instead be functional-aware or use explicit structure in the distribution bias.

6 Conclusions and Next Steps

The main structural conclusion of the project is that the experiments should no longer be presented as a single linear sequence of QAE trials. They form three interlocking workflows.

- **Workflow A** studies angle-structured QAE integration of a function g under uniform quadrature. This is where the original g_0, g_1, g_2 hierarchy, MLAE/IQAE/canonical-QAE compar-

isons, and the forthcoming quadrature-rule experiments belong.

- **Workflow B** studies Grover–Rudolph preparation of a finite probability law p and direct computation of finite-law quantities $I_p(f)$. This includes the distribution-validation campaigns, affine and beta-bump scaling, MC/QMC baselines, the $n = 6$ affine result, and the centered Gaussian test.
- **Workflow C** is the backend-aware control layer. It explains the role of local audits, compilation choices, noise-model simulations, mitigation, layout selection, repetition rules, and rejection of amplified circuits whose predicted or measured behaviour is not stable.

This organization also resolves a conceptual ambiguity in the earlier draft. The calibration functions g_0, g_1, g_2 are used as integrands in Workflow A. The later Grover–Rudolph profiles such as `gr_affine_n4`, `gr_affine_n6`, and `gr_gaussian_mid_sigma025_n6` are used to define probability laws in Workflow B; the quantities of interest are then classical test functions integrated against those laws.

Phase 1 establishes a complete and reproducible first benchmark cycle on IBM hardware. The main conclusions are:

- local audit before QPU execution is essential; it prevented blind escalation to deeper circuits and made the $g_0, k = 2$ anomaly interpretable;
- compilation settings materially affect hardware accuracy, even when the logical and saved transpiled circuits are ideal-correct;
- optimization level 2 provided the accepted evidence chain on `ibm_fez`;
- after compilation tuning, the affine and quadratic benchmarks both pass through $k = 2$ and produce accurate MLAE estimates.

The accepted Phase 1 result set should be treated as the baseline for the next stage. Further QPU spending should pause until Phase 2 is designed explicitly. The most natural Phase 2 directions are:

1. compare layout-selection strategies for the same g_0, g_1, g_2 schedules;
2. add readout mitigation and zero-noise extrapolation to quantify whether the residual $g_0, k = 2$ bias can be reduced;
3. extend the benchmark from the SISC calibration functions to larger structured Grover–Rudolph instances;
4. convert the current report into an article-ready methodology paper, emphasizing anomaly detection, compilation-sensitive correction, and reproducible hardware evidence.

The immediate Phase 2 action is therefore a credit-free layout and compilation audit over optimization levels and transpiler seeds on `ibm_fez`. The audit ranks candidates by backend-noise deviation, two-qubit-gate count, and depth before any additional QPU jobs are considered.

The first Phase 2 audit identified an actionable $g_0, k = 2$ candidate with optimization level 2 and transpiler seed 67890. The candidate has 19 two-qubit gates, depth 76, and backend-noise predicted deviation 0.008545, compared with the accepted Phase 1 hardware deviation 0.047363. This motivates a two-circuit microcampaign repeating only $g_0, k = 0$ and $g_0, k = 2$. The strong g_1 and g_2 Phase 1 results should not be repeated unless mitigation or larger instances are introduced.

The Phase 2 microcampaign did not confirm the audit prediction. The measured $g_0, k = 2$ probability was 0.306641, with absolute deviation 0.056641, compared with the audit prediction

0.258545 and deviation 0.008545. The corresponding MLAE estimate was $\hat{a} = 0.239373$, with absolute error 0.010627. This is slightly worse than the accepted Phase 1 g_0 repeat. The conclusion is that the backend noise model used in the audit is not sufficiently predictive for this sensitive circuit. Further QPU spending on random seed or layout searches is therefore not justified; the next Phase 2 step should be explicit mitigation. The next prepared microcampaign uses Qiskit Runtime dynamical decoupling and Pauli/measurement twirling on only the g_0 calibration pair, keeping the accepted Phase 1 compilation seed fixed so that mitigation is the experimental variable.

The mitigation microcampaign improved the g_0 calibration result. With dynamical decoupling and gate/measurement twirling, $g_0, k = 2$ gave $\hat{p} = 0.291016$, absolute deviation 0.041016, and MLAE estimate $\hat{a} = 0.242160$ with absolute error 0.007840. This is the best g_0 result obtained so far, improving on the accepted Phase 1 repeat 0.008949 and the Phase 2 layout-seed repeat 0.010627. The circuit nevertheless remains in the watch region, so the project should treat g_0 as a sensitive calibration case and use the robust g_1/g_2 results as the main positive hardware evidence.

Phase 1b simulator experiments support MLAE as the main hardware workflow. The IQAE-style adaptive proxy reduces ideal statistical error, but at a substantially larger oracle-query budget. Canonical QAE and MC/QMC are degenerate on the current four-point calibration family, so the next fair comparison should use non-special amplitudes or larger structured Grover–Rudolph instances.

The non-degenerate simulator extensions support the same decision. Across synthetic amplitudes and larger midpoint-grid functions, the IQAE proxy improves ideal MAE by roughly a factor of two, but typically requires about twice the oracle-query budget of MLAE $K = \{0, 1, 2\}$ and introduces adaptive execution overhead. MLAE therefore remains the preferred near-term hardware method.

The project has now reached the next methodological checkpoint: larger Grover–Rudolph state preparation has been validated directly before QAE is attached to it. The Phase 0-GR audit shows that the paper-aligned uniformly-controlled- R_y construction is viable on `ibm_fez` for eight- and sixteen-point dyadic grids. The first hardware distribution campaign then measured `gr_sin2_n3`, `gr_sin2_half_n3`, and `gr_affine_n4`. The two eight-point circuits achieved TVD 0.0313 and 0.0216, while the sixteen-point affine circuit achieved TVD 0.0479. These data support the state-preparation workflow, but also show that the first sixteen-point case is the right object for a focused repeat. That repeat, performed with 4096 shots, dynamical decoupling, and twirling, reduced TVD to 0.0341 and the maximum per-cell deviation to 0.0086. This is sufficient evidence to prepare the first amplified test, but it should remain minimal: `gr_affine_n4`, marked event $x \geq 1/2$, and MLAE schedule $K = \{0, 1\}$. The first amplified readout confirms that this caution was necessary: $k = 0$ remains compatible with the distribution-measurement scale, but $k = 1$ gives $\hat{p}_1 = 0.173828$ instead of the expected 0.052765. The resulting MLAE error is 0.059854. The $k = 2$ amplified circuit should therefore remain deferred. The next action is a credit-free diagnosis of `gr_affine_n4`, $k = 1$, searching for a compilation/layout candidate whose ideal and noisy predictions are consistent before any repeat is submitted. That diagnosis identified an opt-3, seed-12345 candidate with depth 152 and 35 two-qubit gates, compared with depth 225 and 51 two-qubit gates in the executed opt-2 circuit. A single-circuit repeat of $k = 1$ with this candidate is justified; $k = 2$ remains deferred. The repeat improved the hardware probability from 0.173828 to 0.124023, but the remaining deviation, 0.071259, is still too large. The next step should use the existing counts to test readout postprocessing and event-level bias before any further QPU submission. That test has now been performed with the real transpiled event-qubit layouts. The event bit is measured on $q[137]$ in the opt-2 circuit and on $q[134]$ in the opt-3 repeat. Current backend assignment probabilities are below the level needed to explain the observed lower-to-upper leakage; readout correction leaves the opt-3 marked probability essentially unchanged, 0.124023 to 0.124199. Therefore the next investigation should target

circuit-level leakage, amplification sensitivity, and mitigation beyond final readout assignment, while $k = 2$ remains deferred. A subsequent block-level noisy simulation confirms this direction: the Grover–Rudolph loading block A remains stable, but the full $AS_0A^\dagger OA$ iterate moves the event mass from the ideal 0.052765 to 0.112793 under the backend-noise model. The practical next step is therefore to study lower-depth or partially mitigated amplification blocks before any additional hardware campaign. The first variant audit has already ruled out simple replacements of the zero reflection: `mcp`, diagonal, and direct-tree variants are all worse than the current `ucry+mcx` opt-3 candidate. Consequently, further progress likely requires a different amplification strategy or a smaller event/circuit family, not just a different decomposition of the same $k = 1$ Grover iterate.

The constructive alternative is to use the measured Grover–Rudolph distribution directly. For the same `gr_affine_n4` upper-half event, the mitigated distribution estimate gives 0.671387 against the exact value 0.680380, with absolute error 0.008993 and a Wilson interval containing the truth. This is substantially better than the available amplified MLAE estimate. The immediate project decision is therefore to treat direct mitigated Grover–Rudolph distribution estimation as the near-term hardware workflow for sixteen-point grids, while amplified MLAE remains a diagnostic target requiring additional circuit-level stabilization. The functional-level analysis refines this decision: event probabilities and smooth test-function expectations are already useful, whereas right-tail-sensitive moments and payoff-like functions still reveal residual distribution bias. Future hardware campaigns should therefore report quantity-specific error, not only global TVD or QAE amplitude error. Classical postprocessing can quantify this bias, but the first leave-one-out test shows that a single affine correction across test functions is not robust. The next calibration layer should therefore be functional-aware or distribution-structure-aware, rather than a global scalar correction.

The Phase 5 mitigated distribution suite strengthens this decision while adding one caution. The deferred localized profile `gr_beta_bump_n4` is hardware-viable as a direct distribution experiment, with TVD 0.0456 and maximum cell deviation 0.0241, but it is not yet clean enough to attach amplified QAE. The two eight-point repeats show that Runtime mitigation is not uniformly beneficial: it is neutral for `gr_sin2_half_n3` and slightly worsens `gr_sin2_n3`. A follow-up audit and focused repeat of the exact `gr_beta_bump_n4` law, $F(x) = x(1-x)^4$, further supports caution. Although the audit identified an opt-3, seed-45678 candidate with predicted TVD 0.0217, the hardware repeat gave TVD 0.0502. The maximum cell deviation improved from 0.0241 to 0.0170, but errors for quantities of interest were mixed and most did not improve. The next experimental step should therefore not be a deeper Grover schedule. Amplified QAE remains deferred for localized profiles; the productive path is direct distribution diagnostics, quantity-specific error reporting, and model-aware postprocessing.

Phase 6 begins that postprocessing program. The first extended distribution analysis adds classical fidelity, Hellinger distance, Euclidean distance, and bootstrap intervals to TVD. These metrics reinforce the current decision: `gr_affine_n4` benefits from mitigation, `gr_sin2_n3` does not, and the `gr_beta_bump_n4` repeat is not a robust improvement even though its largest cell error is smaller. The simulator-only purification diagnostic inspired by mixed-state preparation confirms that the Grover–Rudolph loader can be viewed as an eigenvalue-encoding layer with an optional entropy-injection register. This is a useful conceptual extension, but not yet a reason to spend QPU credits. The Phase 6 decision table therefore promotes only direct distribution and selected quantity-of-interest estimates: `gr_sin2_half_n3` is a direct distribution result, `gr_sin2_n3` is a direct result with caveats, and mitigated `gr_affine_n4` supports selected functionals. Localized `gr_beta_bump_n4` profiles remain diagnostic only. No Grover–Rudolph profile is currently promoted to amplified QAE on `ibm_fez`. At the functional level, however, the data already contain a set of defensible direct estimates: five case–functional pairs have absolute error below 5×10^{-3} with bootstrap coverage of the exact value, and seven more have error below 10^{-2} with coverage. This gives a clear near-term publication path based on direct

Grover–Rudolph probability loading, metric-rich distribution benchmarking, and quantity-of-interest postprocessing.

The Phase 7 higher-shot repeats close the first hardware campaign. They show that more shots alone do not turn every measured distribution into a stable global probability-preparation claim. Among the final direct-distribution results, only mitigated `gr_affine_n4` is promoted as a paper-quality global distribution result, with TVD 0.028017 and maximum cell deviation 0.008197. The localized `gr_beta_bump_n4` profile remains a backend diagnostic distribution, with TVD 0.057598, despite having two useful functionals below 10^{-2} . The two $n = 3$ raw repeats also reinforce the same distinction: several algebraic functionals remain useful, but the global probability laws are not uniformly stable under repetition.

The final paper-quality set should therefore be stated at two levels. At the distribution level, the defensible hardware claim is the mitigated `gr_affine_n4` loader. At the algebraic-probability level, the strongest functionals are $I_p(\sin(\pi x))$ for `gr_affine_n4` and $I_p(x)$ for `gr_sin2_n3`. A second tier of useful results, including selected call-payoff and moment functionals, should be reported with explicit bias caveats. All rejected or diagnostic entries remain scientifically important: they identify backend drift, localized-profile sensitivity, and event-tail bias, which are precisely the effects that would be hidden by reporting only a single QAE amplitude error.

The first cross-backend Phase 8 test strengthens this conclusion. Repeating the mitigated `gr_affine_n4` direct-distribution campaign on `ibm_kingston` improves the global distribution metrics relative to `ibm_fez`: TVD decreases from 0.028017 to 0.021823, Hellinger distance from 0.023874 to 0.019060, and classical fidelity increases from 0.998860 to 0.999274. Kingston also reduces the absolute error for every selected functional, including the previously diagnostic upper-half event. The upper-half event is still not promoted to the primary paper-quality set, but the systematic improvement confirms that a nontrivial part of the residual error is backend dependent rather than intrinsic to the Grover–Rudolph circuit.

The campaign should now proceed selectively. The affine result is strong enough to serve as the cross-backend anchor for the paper. Further QPU spending should focus on one of two sharply defined questions: whether the $n = 3$ drift seen on `ibm_fez` also appears on `ibm_kingston`, or whether the localized `gr_beta_bump_n4` instability is backend-specific. Amplified QAE should remain deferred until these direct-distribution cross-backend controls are understood.

The next methodological step is Phase 9: integration scaling. The goal is to test whether the direct Grover–Rudolph workflow remains useful when the finite probability law is refined from 2^3 and 2^4 grid cells to 2^5 cells. This phase explicitly separates discretization error, state-loading error, finite-shot error, and backend-noise-induced error for each quantity of interest. This separation is important for the paper: a hardware result should not be interpreted as a quantum integration advantage unless the classical grid error, the probability-loading error, and the quantity-specific hardware bias are all reported. Phase 9 should therefore start as a credit-free audit on `ibm_kingston`, using the affine cross-backend result as the anchor and treating $n = 5$ cases as candidates only if their transpiled depth, two-qubit count, noisy TVD, and maximum quantity-of-interest error remain below the predeclared thresholds.

The first Phase 9 hardware submission confirms that this route is viable. On `ibm_kingston`, the mitigated 32-cell affine law achieves TVD 0.033694, Hellinger distance 0.031298, classical fidelity 0.998042, and maximum cell deviation 0.008438. This promotes `gr_affine_n5` to the first paper-quality scaling result at the distribution level. At the functional level, however, only $I_p(\sin(\pi x))$ is cleanly compatible with the exact finite-law value, with absolute error 0.003601 and bootstrap interval coverage. The call payoff is usable with a bias caveat, while the mean, second moment, and upper-half event remain diagnostic.

The localized 32-cell stress test then establishes the current boundary of the method. Although the `gr_beta_bump_n5` circuit compiles with depth 162 and 44 two-qubit gates, the hard-

ware result gives TVD 0.073298, Hellinger distance 0.110712, classical fidelity 0.975636, and no quantity-of-interest interval coverage. This does not pass the Phase 9 distribution threshold. The conclusion is therefore sharper than a uniform success or failure: direct Grover–Rudolph sampling scales to 32 cells for the affine law, but localized probability profiles remain a backend-sensitive stress regime. Amplified QAE should still remain deferred; the next methodological step should compare this QPU sampling workflow against classical MC/QMC baselines under the same finite laws.

Phase 10 performs that comparison without spending additional QPU credits. It shows that the 32-cell affine result is not uniformly competitive with MC at the same sample count. The strongest finite-law integral remains $I_p(\sin(\pi x))$, where the QPU absolute error 0.003601 is comparable with the MC RMSE 0.003294 for $M = 8192$. The mean, second moment, call payoff, and upper-half mass are instead dominated by hardware bias relative to MC. The localized beta-like law is worse than MC for every tested quantity and should remain a diagnostic stress case. The QMC baseline, implemented through the exact finite inverse CDF, is much more accurate than both QPU and MC once the law is known. This reinforces the right interpretation of the project at this stage: we have demonstrated a reproducible hardware workflow for preparing and sampling structured finite laws, but not a quantum integration advantage over classical sampling baselines.

The angle-structure classification should now be treated as a design variable for the next campaign. The experiments support the distinction between QAE-efficient integration and direct Grover–Rudolph distribution integration. For QAE-efficient integration, the relevant object is the angle map $\Theta_g = 2 \arcsin \sqrt{g}$, and low-degree classes $\mathcal{G}_n^{(d)}$ are the natural first candidates because they correspond to simpler amplitude-oracle structure. For direct distribution integration, high angle degree of the test function f does not prevent estimating $I_p(f)$ from measured frequencies, but it is a warning that the same quantity may be expensive or unstable if recast as an amplified QAE amplitude. The next local phase should therefore classify every proposed probability profile and every proposed test function before backend submission, then choose between direct sampling and amplified QAE according to that structural classification and the MC/QMC baseline scale.

This policy has now been made explicit in Phase 11. The decision table promotes the 32-cell affine law with $f(x) = \sin(\pi x)$ as the current paper-quality direct-sampling result. It also identifies low-angle $\sin^2(\pi x)$ pairs, such as the upper-half event, as simulator-level amplified-QAE candidates, and defers QPU repeats for localized beta-like tail quantities. This is the strongest current use of the angle-structure theory in the experimental workflow: it does not claim that low angle degree alone implies a quantum advantage, but it gives a reproducible pre-submission filter for deciding which integrals are structurally appropriate for amplified QAE and which should remain in the direct Grover–Rudolph sampling regime.

The Phase 12 local QAE-amplitude audit adds a second filter: the scalar amplitude $a = I_p(f)$ and the selected amplification schedule must be informative. The low-angle upper-half tests for the $\sin^2(\pi x)$ profiles have $a = 1/2$, so $p_k(a) = 1/2$ for every Grover level and they are not useful QAE estimation tests despite their favorable angle class. The first meaningful non-degenerate augmented-QAE candidate is instead the eight-cell $\sin^2(\pi x)$ law with $f(x) = \sin(\pi x)$. This pair remains simulator only for now; a backend-aware noisy audit should precede any QPU submission.

That backend-aware audit has now been performed in Phase 13. On both `ibm_kingston` and `ibm_fez`, the noisy model drives the augmented-QAE marked probabilities toward approximately 1/2, even though the ideal values for $k = 0$ and $k = 1$ are well separated. The full $K = \{0, 1, 2\}$ schedule is also not viable under the current cost thresholds. Thus the dual angle-structure workflow has served its intended purpose: it identified a mathematically meaningful QAE candidate, then rejected it before QPU spending once backend-aware noise was included. For the next hardware campaign, the evidence still favors direct Grover–Rudolph distribution

sampling and quantity-of-interest postprocessing over amplified QAE.

Phase 14 begins that next direction by testing direct integration scaling locally from $n = 4$ to $n = 6$. In a local `cz`-basis audit, all three families reach 64 midpoint cells without exceeding the provisional cost thresholds; the $n = 6$ cases require 62 two-qubit gates in the best local transpilation. This makes $n = 6$ a plausible target for backend-aware auditing, but not yet for QPU execution. The Phase 9 and Phase 10 evidence still matters: the affine family is the most credible scaling candidate, while the localized beta-like family should be treated primarily as a stress diagnostic unless a backend noise audit shows a clear improvement.

The backend-aware Phase 14 audit refines this decision. On `ibm_kingston`, all three $n = 6$ candidates pass the pre-submission thresholds, while on `ibm_fez` only the localized beta-like profile passes. The recommended next QPU microcampaign is nevertheless the affine 64-cell law on `ibm_kingston`, because it is the direct continuation of the 32-cell affine result that already passed the Phase 9–10 paper-quality filters. The beta-like 64-cell law should be reserved for a later stress test, not used as the main scaling claim.

The executed 64-cell affine campaign confirms this nuanced view. At the distribution level the TVD rises to 0.058478, close to the Phase 14 acceptance boundary, but the maximum cell deviation remains small. At the integration level the smooth quantity $I_p(\sin(\pi x))$ improves to absolute error 0.000752 and passes interval coverage, making it the strongest current paper-quality direct-integration result. In contrast, the upper-half mass, the mean, the second moment, and the call payoff show clear residual bias. Future claims should therefore be quantity-specific: the experiment supports direct Grover–Rudolph integration for selected smooth quantities on 64 cells, not a uniform backend-independent integration advantage.

The Phase 15 Gaussian test reinforces the same conclusion from the opposite direction. A smoother centered law with a much shallower circuit improves the upper-half mass and the mean, but it worsens $I_p(\sin(\pi x))$ and does not improve the global distribution TVD. The finite-law experiments should therefore be presented as a controlled hardware study of probability-law preparation and quantity-specific integration, not as a single monotone scaling story.

This also clarifies the transition to the next phase. The direct Grover–Rudolph workflow estimates $I_p(f)$ for a prepared law p . The angle-structure paper studies a different numerical-integration object: quadrature-rule QAE for $I[g] = \int_0^1 g(x) dx$, with explicit left, midpoint, right, and Simpson rules. The next experimental block should use the finite-law results as hardware context, but should return to the paper’s quadrature formulation in order to test the encoding-complexity theory directly.

That transition has now started with the Phase 15b midpoint-quadrature QAE microcampaign on `ibm_kingston`. The restored Workflow-A scripts separate $Q_n[g]$, $I[g]$, MLAE estimation error, total error, circuit depth, two-qubit gates, shots, and oracle-query count. The hardware results are encouraging: g_1 gives total error 0.001172, g_2 gives total error 0.000140, and the smooth low-amplitude bump gives total error 0.001568. The shifted affine stress case is less accurate, with error 0.010640, and should remain a watch case. The main conclusion is therefore positive but precise: quadrature-rule MLAE on current IBM hardware is viable for the low-degree calibration hierarchy and at least one smooth low-amplitude stress integral, but backend-aware admission remains necessary before extending to larger grids, Simpson combinations, or non-special affine amplitudes.

The `ibm_aachen` repeat sharpens this point. Although `ibm_aachen` had better average two-qubit and readout calibration figures than `ibm_kingston` at submission time, it did not improve the microcampaign globally. Kingston remains better for g_1 , g_2 , and the shifted affine stress case; Aachen improves only the smooth bump slightly. The large affine failure on Aachen, driven by the $k = 1$ amplified probability, shows that backend choice for QAE cannot be made from aggregate calibration numbers alone. The practical rule for the next phase is to run a backend-aware noisy/layout audit for each proposed amplified circuit and to treat cross-backend repeats as validation, not as a substitute for circuit-level admission tests.

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